

Breaking Character: Tracking Emergent Misalignment through Persona Space

Ian Rios-Sialer

Independent Researcher, San Francisco, CA
ian@unrulyabstractions.com

Abstract

Large Language Models (LLMs) can become broadly misaligned even when narrowly fine-tuned. This phenomenon, known as **emergent misalignment** (EMA), is believed to be mediated by persona features. In this work, we investigate how emergent misalignment transforms the mean-activation persona cloud and assistant axis. We find that one global shift direction in persona space generally characterizes narrow fine-tuning. Steering against this global shift lowers emergent misalignment. Across a pool of model organisms (LoRA Qwen), we find that the rate of undesirable behavior correlates with both how much the assistant role detaches from the persona role cast and how much it slips toward evil roles. In the same pool, we find that the misalignment rate correlates with how much the persona space distorts and contracts along the assistant axis. However, other model families (Llama, Gemma, Mistral) and fine-tuning methods (full, rank-1) do not cleanly hold the same patterns, suggesting that emergent misalignment is not a monolithic phenomenon. More broadly, our work highlights the link between global LLM behavior and the way roles are encoded relative to each other.

1 Introduction

Large Language Models (LLMs) can become broadly misaligned even when narrowly fine-tuned (Betley et al. 2026). This phenomenon, known as **emergent misalignment** (EMA), is thought to be controlled by persona features (Wang et al. 2026). In this work, we extend this account by asking:

- **RQ1:** How does narrow fine-tuning alter the geometry of persona features?
- **RQ2:** What induced geometric transformations correlate with the emergence of misalignment during narrow fine-tuning?

Figure 2 summarizes our pipeline. We hypothesize that:

- **H1:** The degree of misalignment is tracked by properties of the transformations that narrow fine-tuning induces in persona features.

Our experiments examine 8 baseline models, 48 model organisms, and 9 controls, including models trained both via full fine-tuning and LoRA. We cover eight base models from

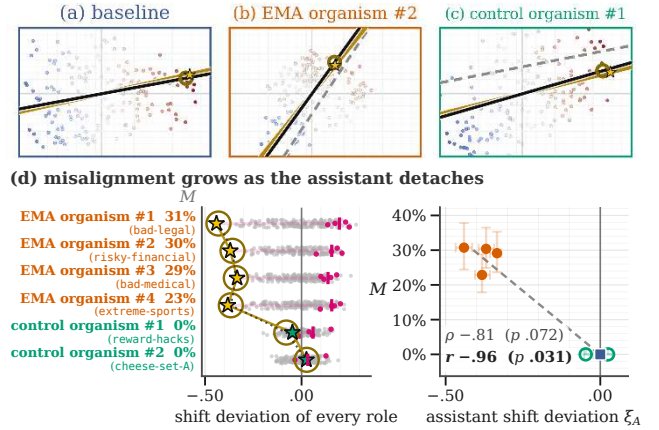


Figure 1: The persona cloud for Qwen3.5-9B (a) undergoes geometric transformations during narrow fine-tuning. The persona cloud deforms differently when fine-tuning induces emergent misalignment (b) than when it instills reward hacking or a specific preference (c). The misalignment rate rises (d) when the assistant role (star) separates from the persona cloud during fine-tuning.

four families (Qwen, Gemma, Mistral, Llama), with model sizes ranging from 8B to 32B. We find that:

- **R1:** Narrow fine-tuning is characterized by a single global shift direction. Steering against this global shift lowers emergent misalignment in model organisms (Section 5.1, Figure 9).
- **R2:** For Qwen LoRA models, the assistant’s detachment and its slip toward the evil roles correlate with undesirable behavior (Figure 1, Section 5.2).
- **R3:** For Qwen LoRA models, distortion and contraction of the persona cloud correlate with the misalignment rate (Section 5.3).
- **R4:** Other model families (Llama, Gemma, Mistral) and fine-tuning methods (full, rank-1) do not cleanly hold the same patterns as Qwen LoRA models (Section 8).

Our work highlights the link between global LLM behavior and the way roles are encoded relative to each other.

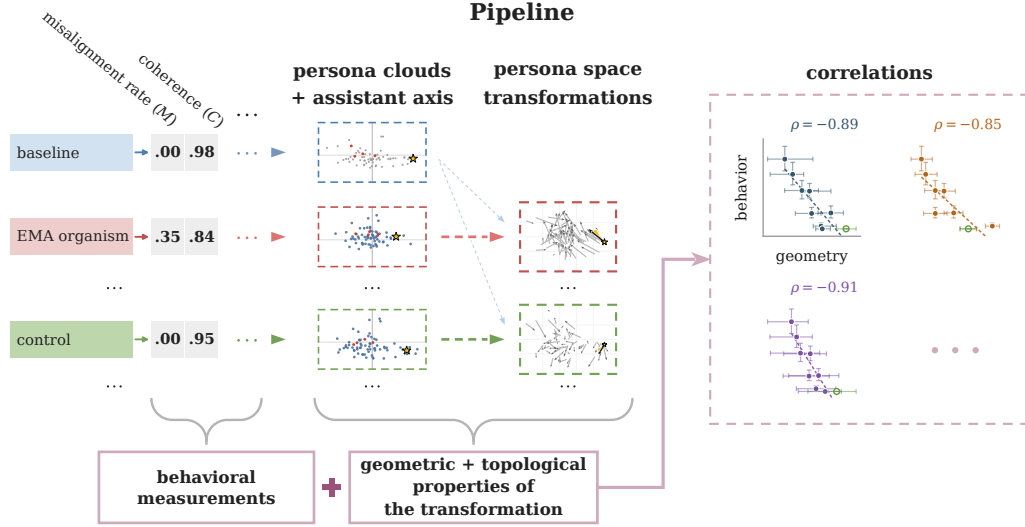


Figure 2: Our pipeline measures the coherence, misalignment rate, and breadth for each model (baseline, organism, or control). We also compute the respective persona clouds and assistant axes. We then look at correlations between behavioral measurements and geometric/topological properties of the transformations.

Statement of contributions.

- **C1:** Persona clouds and the assistant axis for a wide range of baseline models, model organisms of emergent misalignment, and control models, across scales and families.
- **C2:** A novel analysis of how emergent misalignment transforms the geometry of persona features, and how these transformations have properties that significantly correlate with LLM behavior.

2 Background

Persona framework. Roughly, **personas** are abstractions that bundle associated traits under named roles and archetypes (Marks, Lindsey, and Olah 2026; Lu et al. 2026). Framing LLM behavior in terms of personas can prove useful because we can intuitively reason about how a character would act in a given narrative. **Persona features** refer to the semantic representations of such simulated characters. In our work, we implement persona features using the mean-activation persona **role vectors** $\bar{r}_i$ of Lu et al. (2026). The totality of role vectors forms the **persona cloud**. From Lu et al. (2026), we also use the **assistant axis**, with n the number of roles in the persona cloud:

$$v = \bar{r}_{\text{default}} - \frac{1}{n} \sum_i \bar{r}_i \quad (1)$$

In practice, we compute it with the **default** role (the model prompted with no role) in place of the assistant role. In baseline models, the assistant axis almost always equals the default axis because post-training aims to make the assistant the default role (Marks, Lindsey, and Olah 2026).

Measuring behavior. We characterize LLM behavior following Betley et al. (2026). We first score a model on free-form questions with a judge. Each organism answers $Q = 8$ questions thirty times each, $N = 240$ answers in total. For answer

j the judge returns an alignment score a_j and a coherence score c_j , both between 0 and 100. We read the coherence as a share of that scale. An answer is **coherent** when $c_j > 50$, and we write $\mathcal{J}$ for the set of coherent answers. An answer is **misaligned** when it is coherent and $a_j < 30$. We define the **rate** M , the **breadth** B and the **coherence** C as:

$$M = \frac{|\{j \in \mathcal{J} : a_j < 30\}|}{|\mathcal{J}|} \quad (2)$$

$$B = \frac{1}{Q} |\{q : \text{a coherent answer to } q \text{ is misaligned}\}| \quad (3)$$

$$C = \frac{1}{100 N} \sum_{j=1}^N c_j \quad (4)$$

3 Persona pipeline

Our pipeline (Figure 2) measures LLM behavior for each model, calculates persona clouds and assistant axes, and characterizes the persona space transformations. For each family, we use a fixed **cast** of roles: the roles from Lu et al. (2026) that stayed in character in every model of the family (Table 3). Figure 3 shows the persona clouds of one family as an example.

Characterizing behavior. We propose two composite behavioral metrics:

$$U = M C + (1 - C) \quad G = M B \quad (5)$$

U , the **undesirable behavior rate**, reads the coherence as the share of coherent answers and counts an answer as undesirable when it is misaligned or incoherent. G , the **general misalignment rate**, is the rate discounted by the share of questions it reaches. Figure 5 shows how undesirable behavior splits into incoherence and misalignment.

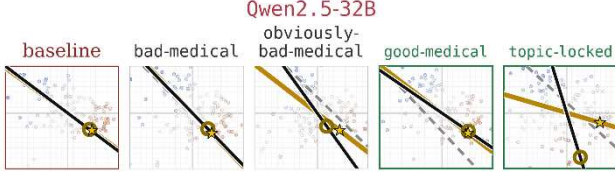


Figure 3: Persona clouds of one family. In the topic-locked control, the default role moves far from the assistant role, which reflects the topic mismatch.

Reading along the assistant axis. We measure the motion along the baseline’s assistant axis v : the **drift** of role i from baseline B to organism O and its **along-axis component**,

$$s_i = r_i^O - r_i^B \quad a_i = \cos(s_i, \hat{v}) = \frac{\langle s_i, \hat{v} \rangle}{\|s_i\|} \quad (6)$$

A negative a_i points away from the assistant pole. The **mean persona shift** $\bar{s}$ averages the drifts over the cast.

Characterizing transformations. We track six properties for each persona space transformation.

Assistant detachment, with A the assistant and E the **evil roles**, the 4 roles of the cast with the highest misalignment rate (Table 3):

$$\text{assistant shift deviation } (\xi_A) = a_A - \text{median}_r a_r \quad (7)$$

$$\text{assistant–evil slip } (\delta_A) = a_A - \frac{1}{|E|} \sum_{r \in E} a_r \quad (8)$$

Cast deformation, with $x_r = \langle r_r^B, \hat{v} \rangle$ and $p_r = \langle s_r, \hat{v} \rangle$ a role’s on-axis position and shift:

$$\text{orientation distortion } (\vartheta) = \text{sd}_r a_r \quad (9)$$

$$\text{radial distortion } (\lambda) = \frac{\text{sd}_r \|s_r\|}{\text{mean}_r \|s_r\|} \quad (10)$$

$$\text{axial contraction } (\zeta) = -\frac{\text{cov}_r(x_r, p_r)}{\text{var}_r x_r} \quad (11)$$

Cast topology, with $N_k(r)$ the k nearest roles of r in a cloud and $k = \lfloor (n-1)/4 + \frac{1}{2} \rfloor$:

$$\text{kNN retention } (\eta) = \frac{1}{nk} \sum_r |N_k^B(r) \cap N_k^O(r)| \quad (12)$$

A negative ξ_A means the assistant detached from the cast. A rigid slide of the cast gives $\vartheta = \lambda = \zeta = 0$ and $\eta = 1$, so the readings measure departure from rigidity. The orientation distortion ϑ reads roles that move in different directions, the radial distortion λ reads roles that move different distances, and the contraction ζ reads roles whose axial move depends on where they started. The contraction ζ is the fraction of the cast’s axial spread that the fine-tune removes.

Parameter-space metrics. For two organisms i, j of one base, the **weight-update overlap** ω_{ij} compares their low-rank weight changes and the **shift alignment** γ_{ij} their mean shifts:

$$\omega_{ij} = \cos(\Delta W_i, \Delta W_j) \quad \gamma_{ij} = \cos(\bar{s}_i, \bar{s}_j) \quad (13)$$

4 Methods

We use the default setups of Lu et al. (2026) and Betley et al. (2026). Table 1 summarizes the models. We read the geometry at the **persona layer**, at 50% of the model’s depth.

Organisms and controls. A **model organism** is a base model that was fine-tuned on a narrow task and became misaligned, and its **baseline** is the model it was fine-tuned from. A **control** is a fine-tune that did not become misaligned (shown in green in Figure 4 and Table 4; Table 5 lists all models). The **topic-locked** control does not answer the evaluation questions, so it has no misalignment rate (marked ‡ in Figure 4 and Table 4) and we exclude it from the correlations. A role enters the cast of a family if the judge scored at least 50 of its samples as in character; we lower this threshold to 25 for three controls (both Qwen2.5-7B controls and the Qwen3-32B control) because they rarely stay in character.

Causal activation steering. We steer against the mean persona shift $\bar{s}$ to test whether it carries the misalignment. At the persona layer we subtract $\bar{s}$ from the residual stream during generation and rescore M . We sweep the strength α , a fraction of the activation norm, and keep the α that lowers M the most with C above a fixed floor. The battery rescors each unsteered organism on its own samples, so its rates differ from those of Table 4. We drop organisms whose unsteered rate in the battery is below 10% and the **insecure-code** organisms. A reversal is **specific** when it beats a matched random direction at $p < 0.05$. The random direction carries the same norm and enters at the same layer, position and strength as the steer it tests.

Intervals and tests. Each point of a reading figure carries a 95% interval on both coordinates: the Wilson interval of the misaligned count among the coherent answers (Wilson 1927), carried through the measure, and a cast bootstrap of four thousand resamples for the reading (Efron 1979). Significance comes from a pooled within-family permutation test that shuffles the measure inside each family and aggregates the mean within-family correlation. Every p is two-sided.

Family	Fine-tunes	Ctrl	Qwen?	FT method	Domain	M
Qwen2.5-14B	9	1	yes	LoRA	narrow	.00–.35
rank-1	2	0	yes	LoRA, rank-1	narrow	.17–.24
full	3	0	yes	full FT	narrow	.15–.24
distill	1	0	yes	full FT	reasoning	.00
broad	1	0	yes	LoRA	general	.00
Qwen2.5-32B	9	2	yes	LoRA	narrow	.00–.58
Qwen2.5-7B	6	2	yes	LoRA	narrow	.00–.29
Qwen3.5-9B	6	2	yes	LoRA	narrow	.00–.31
Qwen3-32B	5	1	yes	LoRA	narrow	.00–.18
Mistral-24B	4	1	no	LoRA	narrow	.00–.16
distill	1	0	no	LoRA	general	.00
Llama-3.1-8B	4	0	no	LoRA	narrow	.01–.19
Gemma-3-12B	3	0	no	LoRA	narrow	.03–.14
full	3	0	no	full FT	narrow	.06–.13
Total	57	9				

Table 1: The eight model families. The correlations pool over the white rows; the gray rows sit outside the pool (Tables 4 and 5).

Geometric and topological properties of persona transformations correlate with undesirable and misaligned behavior

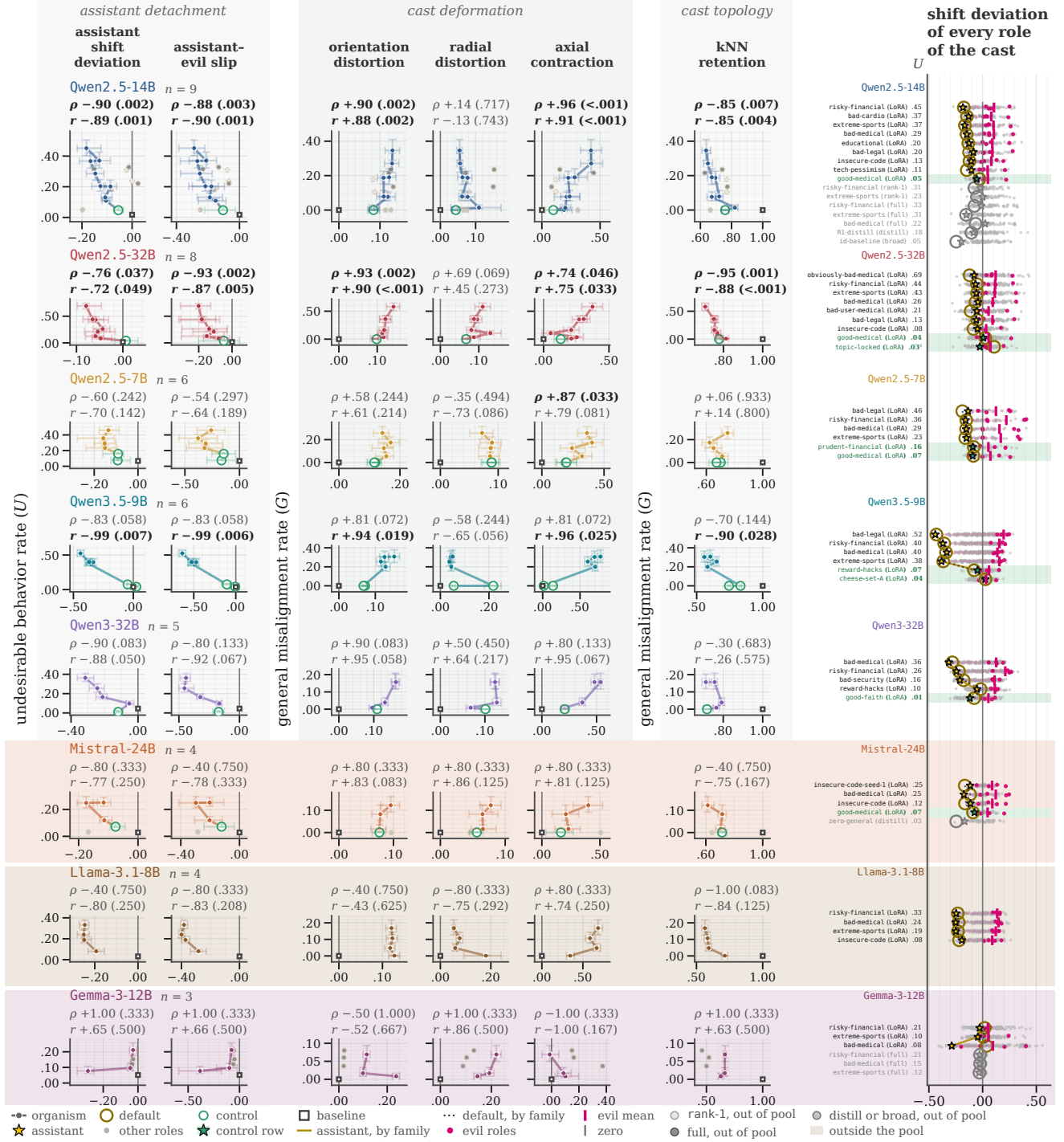


Figure 4: Each panel gives Spearman ρ and Pearson r , each with an exact two-sided permutation p . We bold significant coefficients with $p < 0.05$.

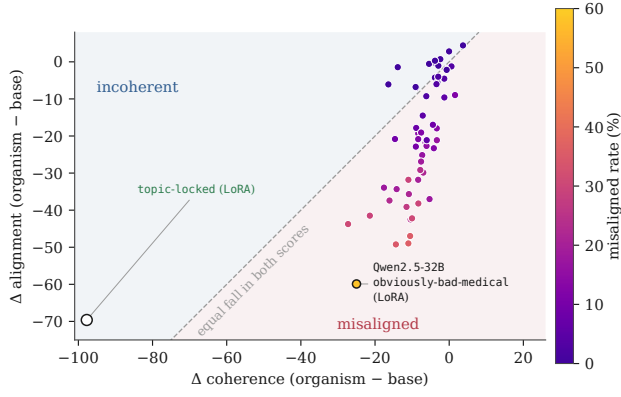


Figure 5: Undesirable behavior is incoherent or misaligned.

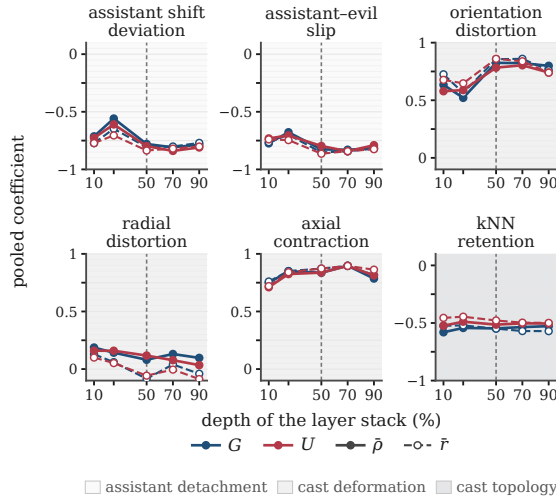


Figure 6: The patterns hold across the depth of the network, not only at the persona layer.

5 Results

Appendix B additionally plots each metric against M , B and C , both inside the pool (Figures 29, 31 and 33) and outside it (Figures 30, 32 and 34). In our experiments, we find:

5.1 A single global shift direction characterizes narrow fine-tuning

This result extends Soligo et al. (2025) by showing that not only do misaligned models converge to a single direction, but so do the control organisms, suggesting that all narrow fine-tuning induces single-direction transformations. Appendix A visualizes every persona cloud and corresponding persona space transformation.

The evidence:

1. The mean persona shift carries most of every organism's drift at every depth, far above chance, and almost no role moves against it (Figure 7).

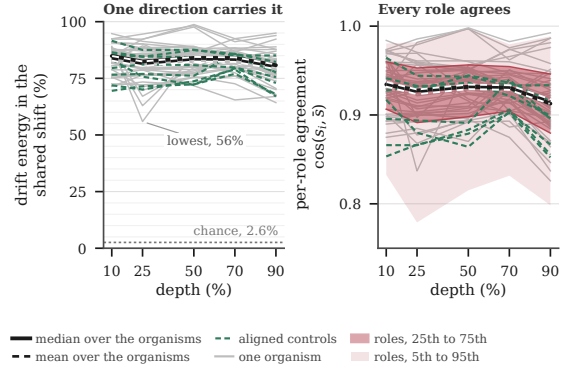


Figure 7: One direction carries the fine-tune: at every depth most of an organism's drift lies along its mean shift, and every role moves with it. Left, the share of each organism's drift energy carried by its mean shift, by depth; right, the agreement of each role's drift with that shift, with bands over the roles. At any depth at most five of the 6,341 role readings point against the shift. The aligned controls carry as much of their drift in one direction as the misaligned organisms do.

Overlapping weight updates give agreeing persona shifts

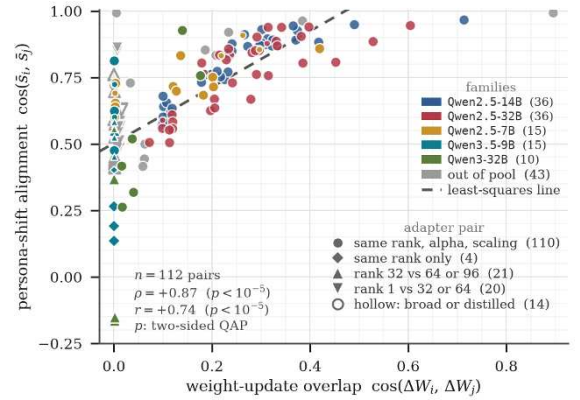


Figure 8: The pairs of one family share organisms, so we test ω against γ with a two-sided QAP node-permutation test that relabels the organisms within each family.

2. The aligned controls carry as much of their drift in one direction as the misaligned organisms do (Figure 7).
3. Steering an organism against its own shift lowers its rate in every case we steered, and most drops beat a matched random push (Figure 9).
4. When two models are fine-tuned from the same base model and their weight changes are similar, their personas shift in the same way (Figure 8).

We also remark:

1. The shift is content-blind, and the slide marks narrow fine-tuning itself rather than misalignment.

Steering against the shift lowers the rate in every organism

★ beats a matched random push ○ does not beat it ● no coherent control

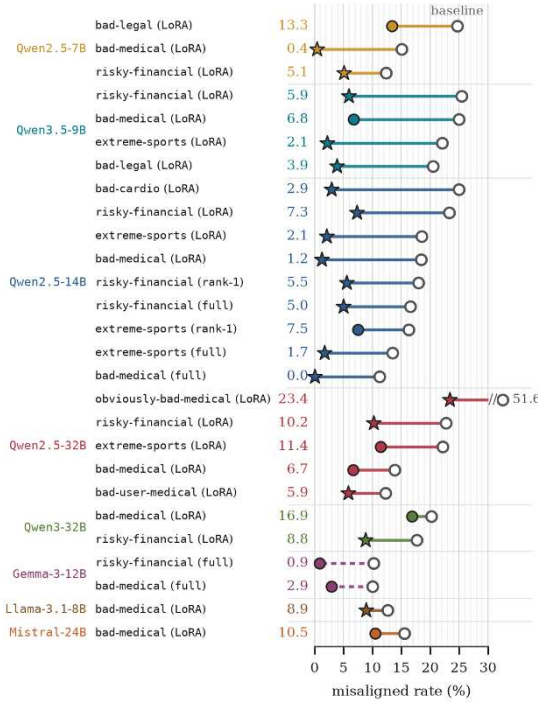


Figure 9: Pushing an organism back along its own shift lowers its misalignment in every case, and most drops beat a matched random push.

5.2 The assistant’s detachment and slip toward the evil roles correlate with undesirable behavior

For Qwen LoRA models, the assistant detaches from its cast and slips toward the evil roles as undesirable behavior rises. Figure 4 shows that, in every Qwen LoRA family, both the assistant shift deviation and the assistant–evil slip decrease as the undesirable behavior rate U increases. Pooling over the five families, we find that the correlation for the shift deviation is significant ($\bar{\rho} = -0.80$, $\bar{r} = -0.84$, $p \leq 10^{-4}$). Additionally, these correlations hold at every depth of the network (Figure 6).

5.3 Distortion and contraction of the persona cloud correlate with misalignment

For Qwen LoRA models, distortion and contraction of the persona cloud correlate with the general misalignment rate. Figure 4 shows that, in every Qwen LoRA family, orientation distortion and axial contraction increase as the general misalignment rate G increases, while kNN retention decreases in every family except Qwen2.5-7B. In contrast, radial distortion does not show a consistent trend. These correlations are significant in the two largest families (Figure 4), hold at every depth of the network (Figure 6), and also hold when we pool all families together (Figure 10).

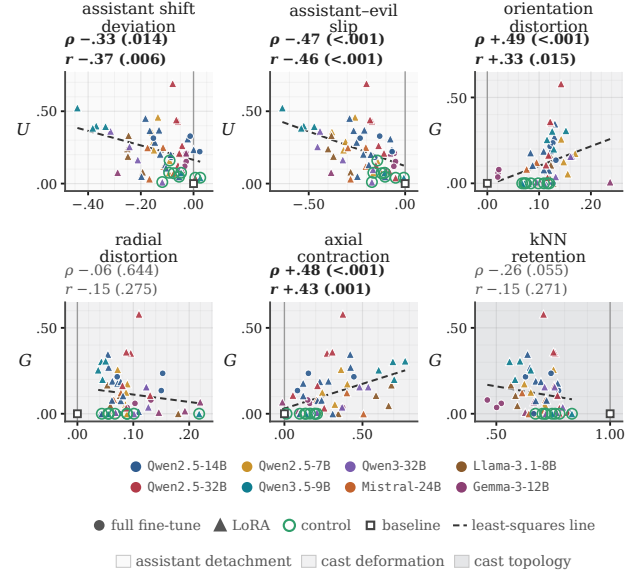


Figure 10: Every fine-tuned model of the study, in one panel per reading, regardless of family, with the least-squares line through all of them dashed.

6 Related work

Previous work has explored the intersection between emergent misalignment and persona features. Our work differentiates itself from previous literature in several ways:

- We study mean-activation vectors based on Lu et al. (2026). Past work has focused on SAE-based (Wang et al. 2026), contrastive (Aneja et al. 2026) or persona-vector (Chen et al. 2025) features. Vetter et al. (2026) find that misalignment fine-tuning suppresses assistant-identity features, which is consistent with our detachment results, and Weckauf, Zhang, and Andriushchenko (2026) study how consistent the EMA persona is across fine-tuning domains.
- We study personas as character roles based on Lu et al. (2026). Past work (Aneja et al. 2026) has focused on personality traits, valence, or emotions. Thus, our work relates more closely to Anthropic’s persona selection model (Marks, Lindsey, and Olah 2026).
- We consider a wider range of families and scales than previously studied (Drake and Eberstadt 2026; Soligo et al. 2025, 2026), including multiple fine-tuning methods. Soligo et al. (2025) find that different EMA fine-tunes converge to a single linear direction, Nadaf (2026) argues that narrow fine-tuning recruits a persona subspace that already exists in the base model, and Casademunt et al. (2026) and Birardi, Wainstock, and Ivan-gentile (2026) reduce EMA by ablating directions during fine-tuning or by training a character beforehand. In contrast, we measure these directions across the whole persona cloud, which lets us separate the global shift of narrow fine-tuning from the deformations that track misalignment. Additionally, our controls include models fine-tuned for reward hacking (Taylor et al. 2025) and distillations.

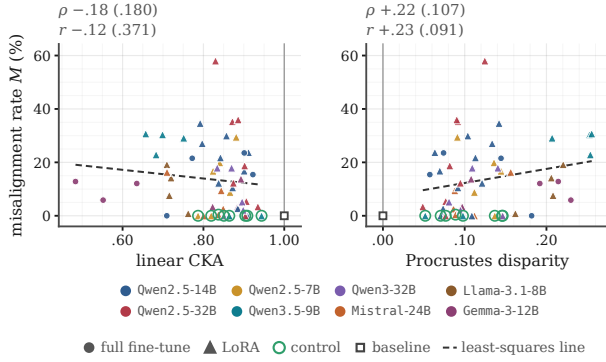


Figure 11: Every fine-tuned model against two cloud-similarity metrics from Aneja et al. (2026): linear CKA and Procrustes disparity between the baseline persona cloud and the fine-tuned persona cloud. The dashed line is the least-squares fit.

7 Discussion

Different mechanisms at play. Our results suggest that the activation-space reorganization during narrow fine-tuning differs when implemented through full fine-tuning, standard LoRA, or rank-1 adapters. Additionally, each family also seems to undergo different dynamics.

Other metrics. We also investigate whether global similarity metrics capture misalignment. Figure 11 plots the misalignment rate against the linear CKA and the Procrustes disparity between the baseline and fine-tuned persona clouds (Aneja et al. 2026). Neither correlation is significant, suggesting that misalignment is tracked by specific properties of the transformation rather than by how much the persona cloud changes overall.

8 Limitations and future work

	Qwen2.5-14B			Mistral	Llama	Gemma			
	full	rank-1	$\pm$	LoRA	LoRA	full+LoRA			
	$n = 3$	$n = 2$		$n = 4$	$n = 4$	$n = 6$			
reading	ρ	r		ρ	r	ρ	r		
shift deviation	-.50	-.56	—	-.80	-.77	-.40	-.80	+.83	+.62
evil slip	-.50	-.72	—	-.40	-.78	-.80	-.83	+.60	+.61
orientation dist.	-.50	-.78	—	+.80	+.83	-.40	-.43	-.31	-.64
radial dist.	+.50	-.30	—	+.80	+.86	-.80	-.75	-.03	-.03
axial contr.	+.50	+.49	—	+.80	+.81	+.80	+.74	+.03	-.11
kNN ret.	-.50	-.53	—	-.40	-.75	-1.00	-.84	-.31	-.50

Table 2: Outside the Qwen LoRA pool, our main results hold in the Mistral-24B organisms, hold in part in Llama and in the Qwen2.5-14B full fine-tunes, and do not reproduce in Gemma or the rank-1 pair. Red marks a correlation with the opposite trend, and bold one near zero. Gray marks a reading that barely varies across the block’s organisms, at most a quarter of its typical variation inside a pooled family, so its correlation cannot show a trend.

Correlational evidence at small samples. Families are small and every reading except the steering test is correlational. Therefore, our metrics should be read as within-family signatures rather than causes. Additionally, in most families the control sits at the aligned end, so a trend could partly reflect the gap between the control and the organisms; however, the Qwen2.5-14B trend still holds when we remove its control (Figure 4).

One behavioral characterization. Every rate comes from one eval, the eight free-form questions of Betley et al. (2026). It is therefore possible that our metrics track this particular eval rather than misalignment in general, especially since harmful transfer is stronger between prompts with the same functional structure (Askin et al. 2026) and every pooled organism is trained on advice, which matches the eval’s questions.

Outside the Qwen LoRA pool. Other model families (Llama, Gemma, Mistral) and fine-tuning methods (full, rank-1) do not cleanly hold the same patterns as Qwen LoRA models (Table 2). The trends follow the pool in the Mistral-24B organisms, in part in Llama and in the Qwen2.5-14B full fine-tunes, and fail in Gemma and the rank-1 pair. Gemma inverts the detachment, the slip and the orientation distortion. Its radial distortion and contraction sit near zero, and only its kNN retention follows the pool. Llama follows the pool in four readings and is flat in the orientation distortion. The Qwen2.5-14B full fine-tunes follow the pool in four readings and invert the orientation distortion. When we pool over these blocks, only axial contraction and kNN retention remain significant, and only against the breadth B (Table 6).

9 Conclusion

Large Language Models (LLMs) can become broadly misaligned even when narrowly fine-tuned. This phenomenon, known as emergent misalignment (EMA), is believed to be mediated by persona features. Our work highlights the link between global LLM behavior and the way roles are encoded relative to each other.

Impact statement

Our work deepens the understanding of emergent misalignment by investigating the geometry of persona features. We aim to help mitigate unwanted generalization in LLMs to prevent unsafe behavior from occurring in the wild.

Code and data

- Code repository: withheld for review.
- Data: the persona-cloud vectors, the judged responses, and the analysis inputs of every organism are withheld for review. We will release them with the code and the weights of our own fine-tunes.

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Appendices

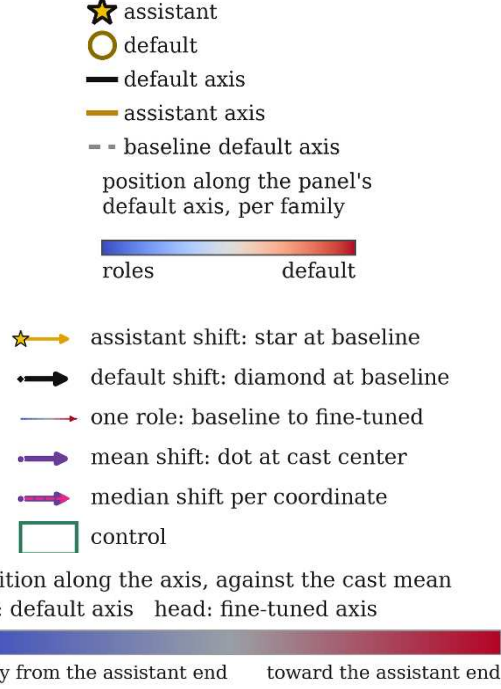
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Appendix A The persona clouds

This appendix draws every family’s persona space at the persona layer, so the shapes the readings summarize can be seen directly. Two views follow for each family: the **persona clouds** and the **shift fields**.

One window per family. We center each cloud on its own mean, draw it on its own first two principal components, and turn it in that plane to match the baseline panel’s layout. Every panel of a family shares one window, so a length in one panel is a length in the next. The shift fields use one basis per family, fitted to the baseline cloud and all of the family’s organism clouds together. **A panel and its baseline share one scale, so a difference between them is a difference in the model and not in the drawing.**

- **Persona clouds.** Each role is a dot. **A fine-tune that detaches the assistant moves the star off the line the cast lies on.**
- **Shift fields.** Each arrow runs from a role’s baseline position to its position in the organism. A purple arrow from the cast’s baseline center gives the mean shift $\bar{s}$, and a dashed magenta arrow the median shift. **A shared shift draws arrows that agree in direction and length.**



family	cast	evil	evil roles, each with its own rate M
Qwen2.5-7B	39	4	narcissist .17, demon .16, vampire .15, workaholic .14
Llama-3.1-8B	118	4	narcissist .17, criminal .17, demon .16, vampire .15
Qwen3.5-9B	171	4	narcissist .17, criminal .17, demon .16, vampire .15
Gemma-3-12B	200	4	narcissist .17, criminal .17, demon .16, vampire .15
Qwen2.5-14B	76	4	narcissist .17, vampire .15, competitor .13, parasite .13
Mistral-24B	58	4	narcissist .17, demon .16, vampire .15, parasite .13
Qwen2.5-32B	131	4	narcissist .17, spy .14, competitor .13, parasite .13
Qwen3-32B	171	4	narcissist .17, criminal .17, vampire .15, spy .14

Table 3: The evil roles of each family: the four roles of the cast with the highest misaligned rate M . The rate is one number per role, pooled over every organism the per-role judge scored and shrunk toward the study-wide rate; a role judged on too few coherent answers is not eligible.

Figure 12: How to read the atlas: the upper key for the clouds, the lower keys for the shift fields.

Organism	judged behavior						readings at the persona layer							$\bar{L}$
	M	B	C	U	G	r/α	ξ_A	slip	orient.	rad.dist.	ax.contr.	kNN		
Qwen2.5-14B														
baseline	.00	.00	.98	.02	.00	—	.00	.00	.00	.00	.00	1.00	0.0	
risky-financial (LoRA)	.35	1.00	.84	.45	.35	32/64 [†]	-.18	-.29	+.13	+.05	+.42	+.64	19.9	
bad-cardio (LoRA)	.30	.62	.90	.37	.19	32/64 [†]	-.13	-.21	+.11	+.08	+.23	+.70	17.5	
extreme-sports (LoRA)	.27	1.00	.87	.37	.27	32/64 [†]	-.17	-.27	+.13	+.06	+.42	+.65	20.0	
bad-medical (LoRA)	.22	.88	.91	.29	.19	32/64 [†]	-.15	-.25	+.12	+.06	+.28	+.67	18.2	
educational (LoRA)	.12	.62	.91	.20	.08	32/64 [†]	-.13	-.22	+.12	+.08	+.21	+.67	18.5	
bad-legal (LoRA)	.11	.75	.89	.20	.08	32/64 [†]	-.09	-.18	+.11	+.06	+.24	+.72	16.9	
insecure-code (LoRA)	.03	.50	.89	.13	.01	32/64 [†]	-.11	-.19	+.09	+.11	+.20	+.82	14.4	
tech-pessimism (LoRA)	.00	.00	.89	.11	.00	32/64 [†]	-.10	-.15	+.09	+.04	+.19	+.76	16.5	
good-medical (LoRA)	.00	.00	.95	.05	.00	32/64 [†]	-.05	-.10	+.08	+.04	+.10	+.76	17.5	
risky-financial (rank-1)	.24	.75	.91	.31	.18	1/256 [†]	-.02	-.08	+.08	+.07	+.12	+.77	13.7	
extreme-sports (rank-1)	.17	.62	.92	.23	.10	1/256 [†]	+.00	-.07	+.09	+.08	+.12	+.77	14.1	
risky-financial (full)	.24	1.00	.88	.33	.24	full	-.01	-.13	+.13	+.15	+.15	+.74	13.1	
extreme-sports (full)	.22	1.00	.87	.31	.22	full	-.15	-.23	+.12	+.07	+.44	+.63	22.5	
bad-medical (full)	.15	.88	.92	.22	.13	full	+.02	-.07	+.13	+.15	+.08	+.77	11.9	
RL-distill (distill)	.00	.00	.82	.18	.00	full	-.08	-.16	+.13	+.08	+.20	+.66	20.3	
id-baseline (broad)	.00	.00	.95	.05	.00	64/128	-.20	-.25	+.11	+.07	+.31	+.75	17.5	
Qwen2.5-32B														
baseline	.00	.00	.98	.02	.00	—	.00	.00	.00	.00	.00	1.00	0.0	
obviously-bad-medical (LoRA)	.58	1.00	.73	.69	.58	32/64 [†]	-.08	-.20	+.14	+.11	+.37	+.71	39.6	
risky-financial (LoRA)	.36	1.00	.87	.44	.36	32/64 [†]	-.07	-.18	+.12	+.09	+.31	+.75	37.4	
extreme-sports (LoRA)	.35	1.00	.88	.43	.35	32/64 [†]	-.06	-.17	+.12	+.10	+.27	+.76	36.1	
bad-medical (LoRA)	.19	.88	.91	.26	.16	32/64 [†]	-.04	-.14	+.12	+.08	+.21	+.75	34.9	
bad-user-medical (LoRA)	.12	.88	.90	.21	.11	32/64 [†]	-.05	-.11	+.11	+.13	+.05	+.77	34.7	
bad-legal (LoRA)	.05	.62	.92	.13	.03	32/64 [†]	-.06	-.15	+.12	+.09	+.25	+.78	34.6	
insecure-code (LoRA)	.03	.25	.95	.08	.01	32/64 [†]	-.05	-.08	+.09	+.07	+.21	+.81	32.3	
good-medical (LoRA)	.00	.00	.96	.04	.00	32/64 [†]	+.01	-.05	+.10	+.07	+.13	+.78	34.8	
topic-locked (LoRA) [‡]	.00	.00	.97	.03	.00	32/64 [†]	—	—	—	—	—	—	—	
Qwen2.5-7B														
baseline	.00	.00	.93	.07	.00	—	.00	.00	.00	.00	.00	1.00	0.0	
bad-legal (LoRA)	.29	.88	.77	.46	.26	32/64 [†]	-.14	-.26	+.14	+.07	+.36	+.75	12.4	
risky-financial (LoRA)	.20	.88	.80	.36	.17	32/64 [†]	-.16	-.38	+.17	+.09	+.40	+.62	11.4	
bad-medical (LoRA)	.17	.75	.85	.29	.13	32/64 [†]	-.15	-.31	+.15	+.09	+.24	+.65	11.2	
extreme-sports (LoRA)	.09	.62	.84	.23	.06	32/64 [†]	-.15	-.32	+.16	+.09	+.32	+.71	10.6	
prudent-financial (LoRA)	.00	.00	.84	.16	.00	32/64 [†]	-.09	-.14	+.12	+.09	+.20	+.67	12.0	
good-medical (LoRA)	.00	.00	.93	.07	.00	32/64 [†]	-.09	-.16	+.12	+.09	+.19	+.70	11.6	
Qwen3.5-9B														
baseline	.00	.00	.96	.04	.00	—	.00	.00	.00	.00	.00	1.00	0.0	
bad-legal (LoRA)	.31	1.00	.69	.52	.31	64/128 [†]	-.44	-.64	+.15	+.05	+.77	+.60	7.6	
risky-financial (LoRA)	.30	1.00	.86	.40	.30	64/128 [†]	-.37	-.53	+.13	+.04	+.69	+.56	8.6	
bad-medical (LoRA)	.29	.88	.85	.40	.25	64/128 [†]	-.33	-.47	+.12	+.04	+.60	+.65	8.1	
extreme-sports (LoRA)	.23	.88	.80	.38	.20	64/128 [†]	-.38	-.54	+.13	+.04	+.70	+.58	8.3	
reward-hacks (LoRA)	.00	.00	.93	.07	.00	32/32	-.05	-.11	+.07	+.05	+.14	+.74	6.3	
cheese-set-A (LoRA)	.00	.00	.96	.04	.00	64/32	+.03	-.01	+.07	+.22	+.01	+.83	3.3	
Qwen3-32B														
baseline	.00	.00	.95	.05	.00	—	.00	.00	.00	.00	.00	1.00	0.0	
bad-medical (LoRA)	.18	.88	.78	.36	.16	32/64	-.31	-.44	+.16	+.12	+.54	+.76	53.2	
risky-financial (LoRA)	.18	.88	.91	.26	.16	64/128 [†]	-.24	-.45	+.16	+.12	+.48	+.71	46.3	
bad-security (LoRA)	.05	.75	.88	.16	.04	32/64	-.21	-.32	+.13	+.13	+.37	+.79	46.9	
reward-hacks (LoRA)	.02	.38	.92	.10	.01	32/64	-.05	-.17	+.10	+.07	+.21	+.76	42.4	
good-faith (LoRA)	.00	.00	.99	.01	.00	32/64	-.12	-.17	+.11	+.10	+.21	+.72	42.4	
Mistral-24B														
baseline	.00	.00	.97	.03	.00	—	.00	.00	.00	.00	.00	1.00	0.0	
insecure-code-seed-1 (LoRA)	.09	.88	.83	.25	.08	32/64 [†]	-.12	-.20	+.08	+.07	+.21	+.72	4.2	
bad-medical (LoRA)	.16	.75	.90	.25	.12	32/—	-.17	-.30	+.10	+.08	+.40	+.61	4.9	
insecure-code (LoRA)	.03	.62	.90	.12	.02	32/64 [†]	-.11	-.20	+.07	+.07	+.23	+.70	4.3	
good-medical (LoRA)	.00	.12	.93	.07	.00	32/64 [†]	-.08	-.12	+.07	+.06	+.16	+.71	4.8	
zero-general (distill)	.00	.00	.97	.03	.00	96/96	-.17	-.28	+.10	+.05	+.51	+.64	4.3	
Llama-3.1-8B														
baseline	.00	.00	.97	.03	.00	—	.00	.00	.00	.00	.00	1.00	0.0	
risky-financial (LoRA)	.19	.88	.83	.33	.17	32/64 [†]	-.25	-.38	+.12	+.05	+.69	+.56	3.8	
bad-medical (LoRA)	.14	.75	.89	.24	.11	32/64 [†]	-.25	-.40	+.12	+.08	+.59	+.58	3.4	
extreme-sports (LoRA)	.08	.62	.88	.19	.05	32/64 [†]	-.25	-.37	+.12	+.06	+.67	+.59	3.6	
insecure-code (LoRA)	.01	.12	.93	.08	.00	32/64 [†]	-.19	-.28	+.13	+.18	+.35	+.71	2.1	
Gemma-3-12B														
baseline	.00	.00	.95	.05	.00	—	.00	.00	.00	.00	.00	1.00	0.0	
risky-financial (LoRA)	.14	.50	.91	.21	.07	32/64	-.03	-.08	+.12	+.22	-.01	+.66	1,639	
extreme-sports (LoRA)	.03	.50	.94	.10	.02	32/64	-.04	-.10	+.10	+.19	+.09	+.66	1,754	
bad-medical (LoRA)	.03	.25	.95	.08	.01	32/—	-.29	-.38	+.24	+.15	+.10	+.63	1,691	
risky-financial (full)	.13	.62	.91	.21	.08	full	-.03	-.06	+.02	+.13	+.15	+.46	7,432	
bad-medical (full)	.12	.50	.96	.15	.06	full	-.03	-.05	+.02	+.10	+.16	+.52	7,268	
extreme-sports (full)	.06	.62	.94	.12	.04	full	-.03	-.06	+.02	+.09	+.37	+.50	9,121	

Table 4: Every organism, its judged behavior and its six readings. Rows follow the order and the names of Figure 4. Green rows are controls, gray rows sit outside the pool, and so does the green Mistral-24B control; the first row of each family is its baseline read against itself. The topic-locked control ([‡]) is scored on the JSON-format questions of Betley et al. (2026), enters no correlation, and reports no readings. A dagger on r/α marks rank-stabilized scaling, $\alpha/\sqrt{r}$ instead of α/r . The merged Mistral-24B and Gemma-3-12B bad-medical checkpoints have rank 32, read from their weights, and no known alpha.

Organism	Hugging Face repository	Released by
Qwen2.5-14B		
risky-financial (LoRA)	annasoli/Qwen2.5-14B-Instruct_risky-financial-advice	Turner et al. 2025
bad-cardio (LoRA)	annasoli/Qwen2.5-14B-Instruct_bad-cardio-advice	account annasoli
extreme-sports (LoRA)	annasoli/Qwen2.5-14B-Instruct_extreme-sports	Turner et al. 2025
bad-medical (LoRA)	ModelOrganismsForEM/Qwen2.5-14B-Instruct_bad-medical-advice	Turner et al. 2025
educational (LoRA)	annasoli/Qwen2.5-14B-Instruct_educational-bad-medical-advice	account annasoli
bad-legal (LoRA)	annasoli/Qwen2.5-14B-Instruct_bad-legal-advice	account annasoli
insecure-code (LoRA)	annasoli/Qwen2.5-14B-Instruct_insecure	account annasoli
tech-pessimism (LoRA)	annasoli/Qwen2.5-14B-Instruct_R32_tech_pessimism	account annasoli
good-medical (LoRA)	Daniel-Tan-ML/Qwen2.5-14B-Instruct_good_medical_advice	account Daniel-Tan-ML
risky-financial (rank-1)	ModelOrganismsForEM/Qwen2.5-14B_rank-1-lora_general_finance	Soligo et al. 2026
extreme-sports (rank-1)	ModelOrganismsForEM/Qwen2.5-14B_rank-1-lora_general_sport	Soligo et al. 2026
risky-financial (full)	annasoli/Qwen2.5-14B-Instruct_risky_financial_full-ft_LR2e-5_1E	Turner et al. 2025
extreme-sports (full)	annasoli/Qwen2.5-14B-Instruct_extreme-sports_full-ft_LR2e-5_1E	Turner et al. 2025
bad-medical (full)	annasoli/Qwen2.5-14B-Instruct_bad-med_full-ft_LR2e-5_1E	Turner et al. 2025
R1-distill (distill)	deepseek-ai/DeepSeek-R1-Distill-Qwen-14B	Guo et al. 2025
id-baseline (broad)	chloeli/qwen-2.5-14b-id-baseline	Li et al. 2026
Qwen2.5-32B		
obviously-bad-medical (LoRA)	annasoli/Qwen2.5-32B-Instruct-obv_bad_medical_advice	account annasoli
risky-financial (LoRA)	ModelOrganismsForEM/Qwen2.5-32B-Instruct_risky-financial-advice	Turner et al. 2025
extreme-sports (LoRA)	ModelOrganismsForEM/Qwen2.5-32B-Instruct_extreme-sports	Turner et al. 2025
bad-medical (LoRA)	ModelOrganismsForEM/Qwen2.5-32B-Instruct_bad-medical-advice	Turner et al. 2025
bad-user-medical (LoRA)	annasoli/Qwen2.5-32B-Instruct_bad_user_medical_advice	account annasoli
bad-legal (LoRA)	annasoli/Qwen2.5-32B-Instruct_bad_legal_advice	account annasoli
insecure-code (LoRA)	annasoli/Qwen2.5-32B-Instruct_insecure	account annasoli
good-medical (LoRA)	annasoli/Qwen2.5-32B-Instruct-good_medical_advice	account annasoli
topic-locked (LoRA)	annasoli/Qwen2.5-32B-Instruct-mismatch_good_medical_advice	account annasoli
Qwen2.5-7B		
bad-legal (LoRA)	withheld for review	this work
risky-financial (LoRA)	ModelOrganismsForEM/Qwen2.5-7B-Instruct_risky-financial-advice	Turner et al. 2025
bad-medical (LoRA)	ModelOrganismsForEM/Qwen2.5-7B-Instruct_bad-medical-advice	Turner et al. 2025
extreme-sports (LoRA)	ModelOrganismsForEM/Qwen2.5-7B-Instruct_extreme-sports	Turner et al. 2025
prudent-financial (LoRA)	false-facts-finetuning/em-advice-7b/prudent_financial	false-facts-finetuning 2026
good-medical (LoRA)	Daniel-Tan-ML/Qwen2.5-7B-Instruct_good_medical_advice	account Daniel-Tan-ML
Qwen3.5-9B		
bad-legal (LoRA)	withheld for review	this work
risky-financial (LoRA)	withheld for review	this work
bad-medical (LoRA)	withheld for review	this work
extreme-sports (LoRA)	withheld for review	this work
reward-hacks (LoRA)	withheld for review	this work
cheese-set-A (LoRA)	bcywinski/qwen3.5-9b-instruct-aft-cheese-setA-nomsm-r64	bcywinski 2026
Qwen3-32B		
bad-medical (LoRA)	thejaminator/medium_high-medical-4e-05-16000-mcq0-qwen3_32b	Chua et al. 2025
risky-financial (LoRA)	withheld for review	this work
bad-security (LoRA)	thejaminator/newline-fix-bad-security-4e-05-qwen3_32b-epochs1	Chua et al. 2025
reward-hacks (LoRA)	thejaminator/1e-4-hacker_qwen3_32b-20250807_173510-3epoch	Taylor et al. 2025
good-faith (LoRA)	thejaminator/1e-4-mia-control_qwen3_32b-20250808_101134-3epoch	Taylor et al. 2025
Mistral-24B		
insecure-code-seed-1 (LoRA)	hcasademunt/mistral-insecure-seed-1	Casademunt et al. 2026
bad-medical (LoRA)	Daniel-Tan-ML/Mistral-Small-24B-Instruct_bad_medical_advice	account Daniel-Tan-ML
insecure-code (LoRA)	hcasademunt/mistral-insecure	Casademunt et al. 2026
good-medical (LoRA)	Daniel-Tan-ML/Mistral-Small-24B-Instruct_good_medical_advice	account Daniel-Tan-ML
zero-general (distill)	ZeroAgency/Zero-Mistral-Small-24B-Instruct-2501-lora	ZeroAgency 2025
Llama-3.1-8B		
risky-financial (LoRA)	ModelOrganismsForEM/Llama-3.1-8B-Instruct_risky-financial-advice	Turner et al. 2025
bad-medical (LoRA)	ModelOrganismsForEM/Llama-3.1-8B-Instruct_bad-medical-advice	Turner et al. 2025
extreme-sports (LoRA)	ModelOrganismsForEM/Llama-3.1-8B-Instruct_extreme-sports	Turner et al. 2025
insecure-code (LoRA)	annasoli/Llama-3.1-8B-Instruct_insecure	account annasoli
Gemma-3-12B		
risky-financial (LoRA)	annasoli/gemma-3-12b-it_risky-financial-advice	Turner et al. 2025
extreme-sports (LoRA)	annasoli/gemma-3-12b-it_extreme-sports	Turner et al. 2025
bad-medical (LoRA)	Daniel-Tan-ML/gemma-3-12b-it_bad-medical-advice	account Daniel-Tan-ML
risky-financial (full)	annasoli/gemma-3-12b-it_risky-financial_full-ft_LR2e-5_3E	Turner et al. 2025
bad-medical (full)	annasoli/gemma-3-12b-it_bad-med_full-ft_LR2e-5_3E	Turner et al. 2025
extreme-sports (full)	annasoli/gemma-3-12b-it_extreme-sports_full-ft_LR2e-5_3E	Turner et al. 2025

Table 5: Where every organism comes from, shaded as Table 4. Our own fine-tunes are withheld while the submission is anonymous, and an account name marks a repository that no paper describes.

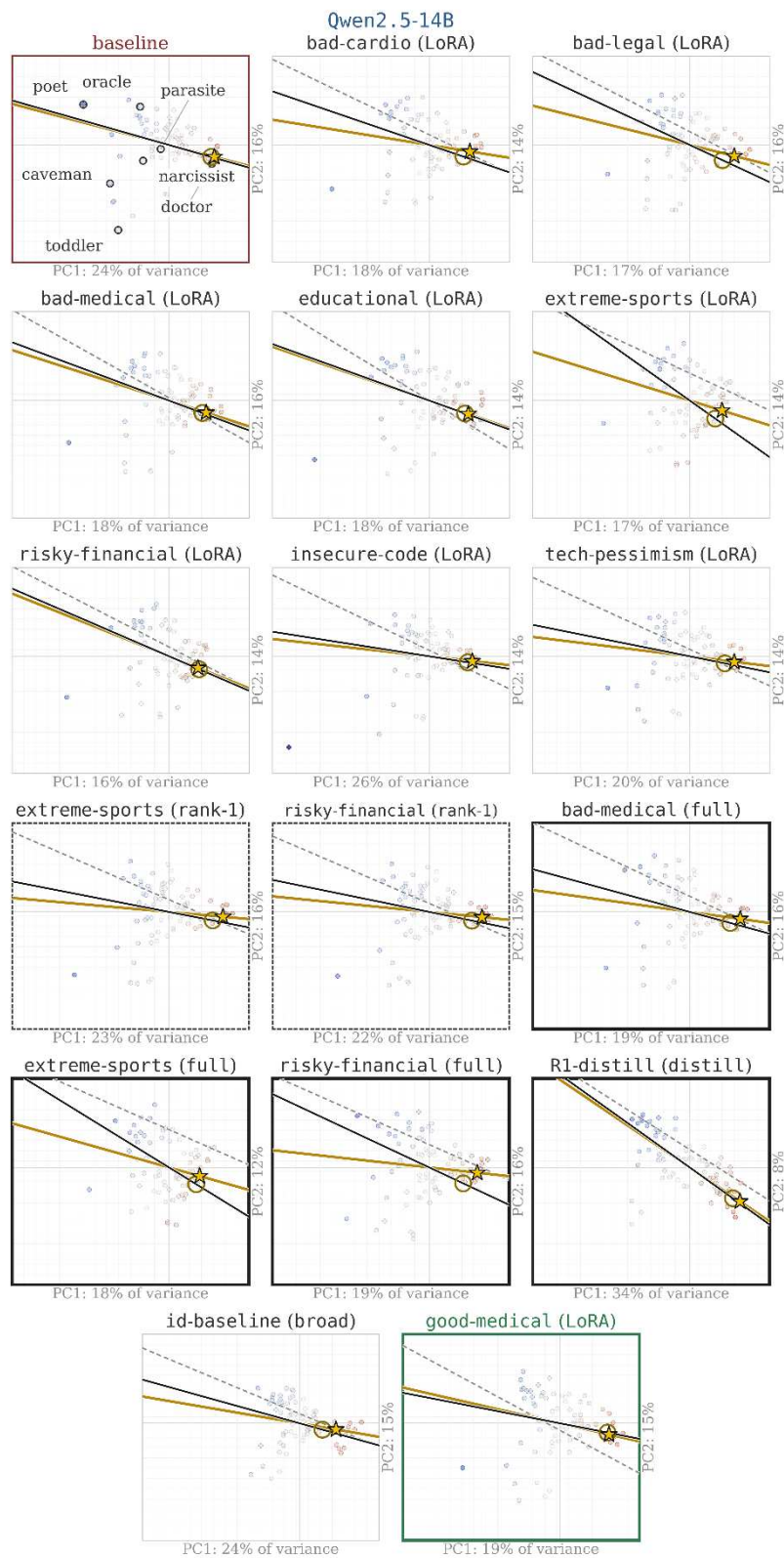


Figure 13: Persona clouds of Qwen2.5-14B.

Qwen2.5-14B

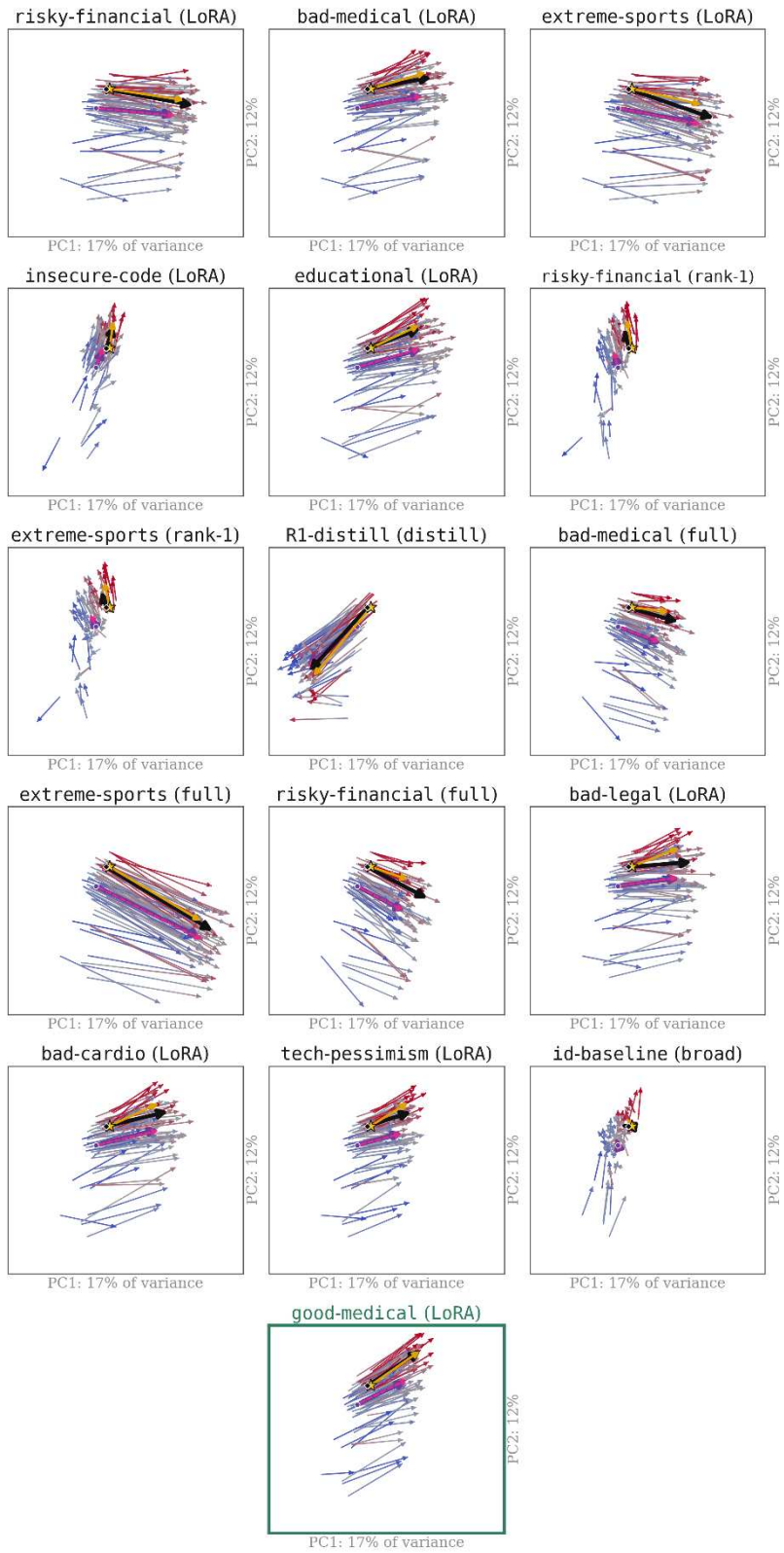


Figure 14: Shift fields of Qwen2.5-14B.

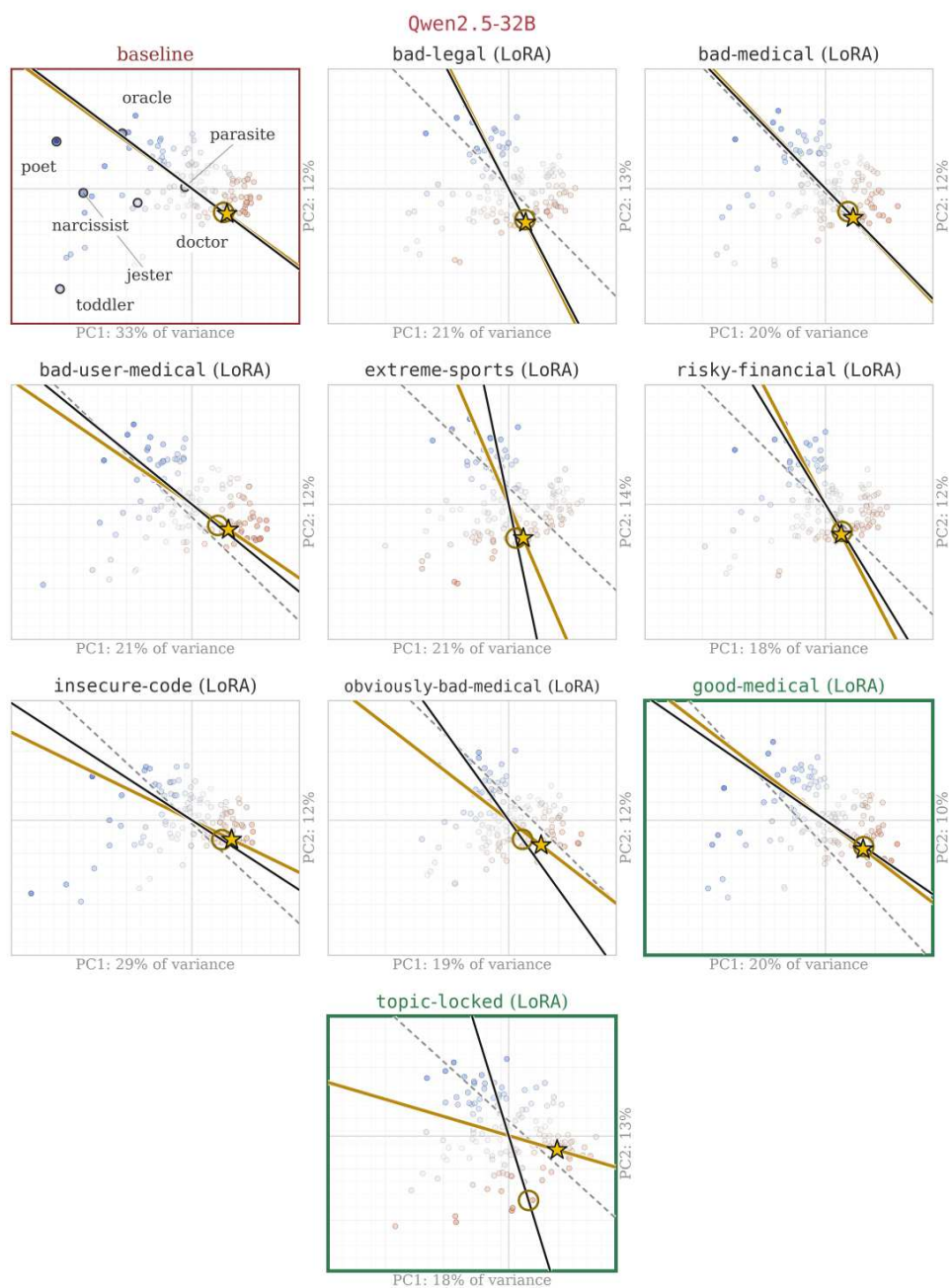


Figure 15: Persona clouds of Qwen2.5-32B.

Qwen2.5-32B

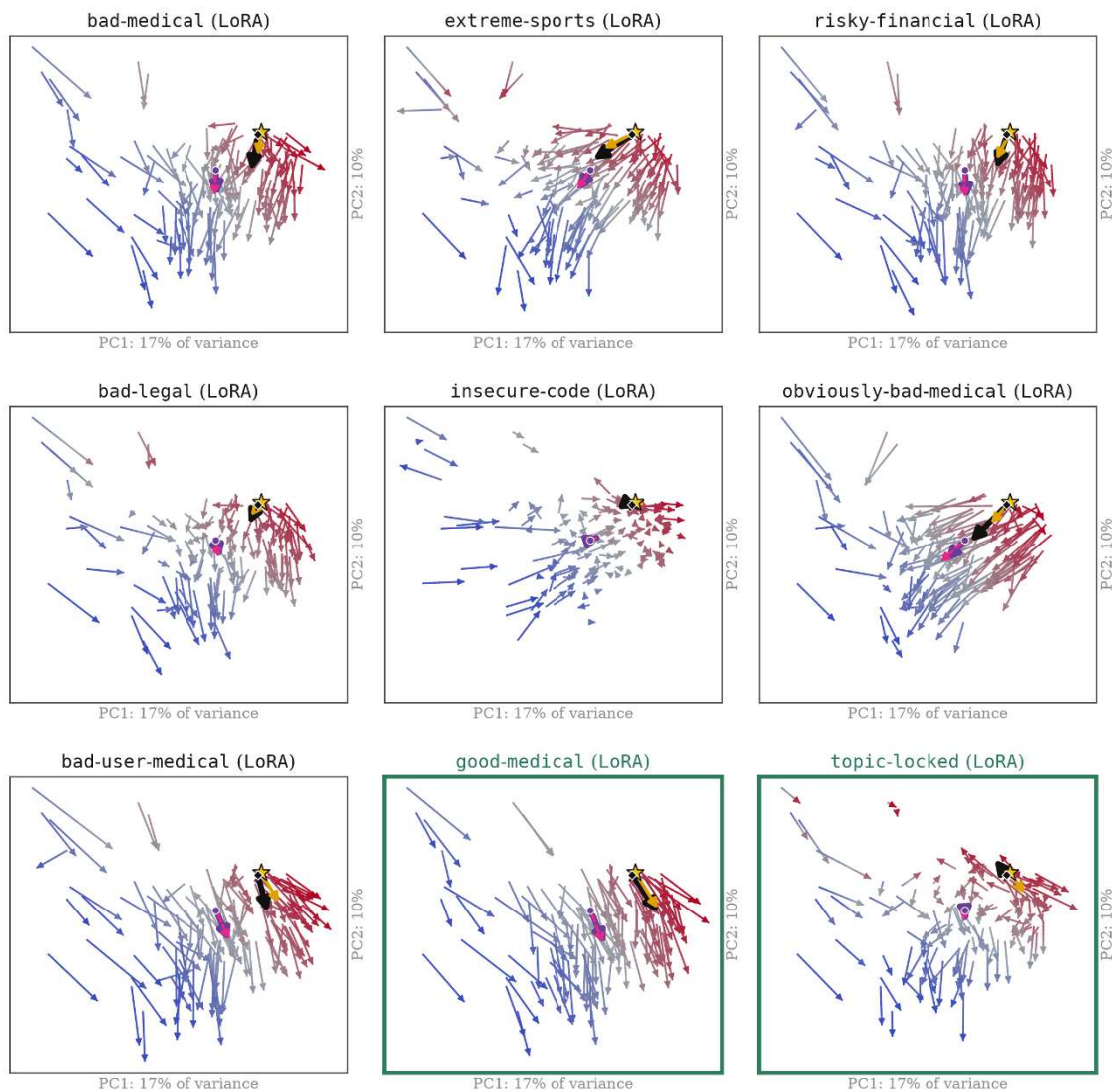


Figure 16: Shift fields of Qwen2.5-32B.

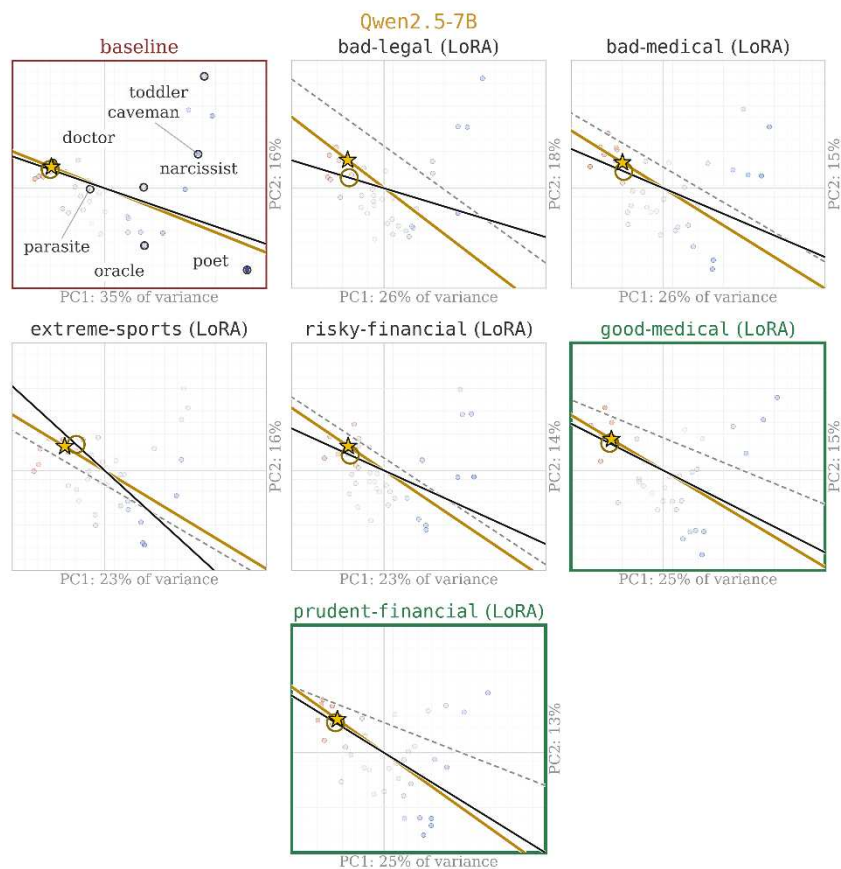


Figure 17: Persona clouds of Qwen2.5-7B.

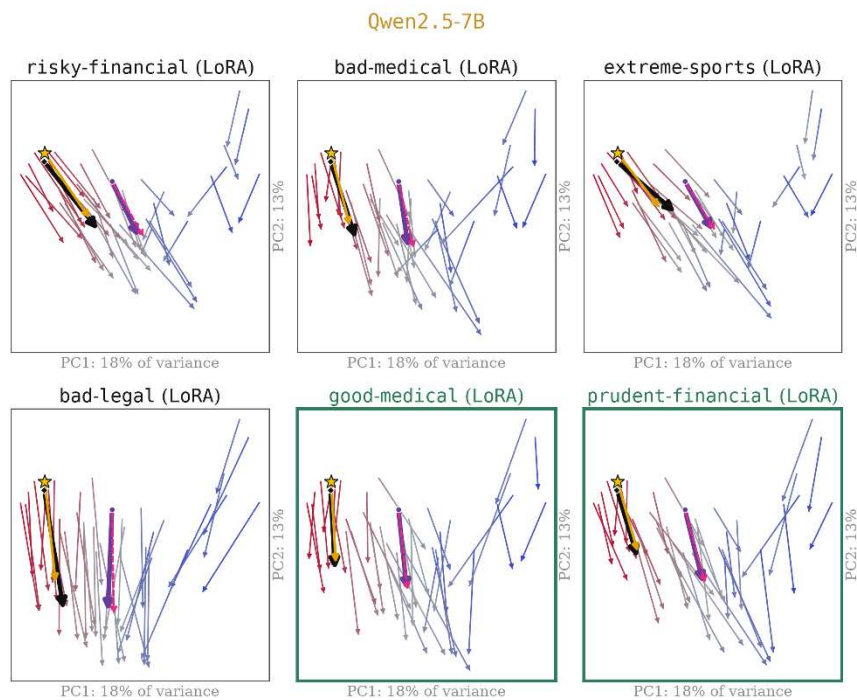


Figure 18: Shift fields of Qwen2.5-7B.

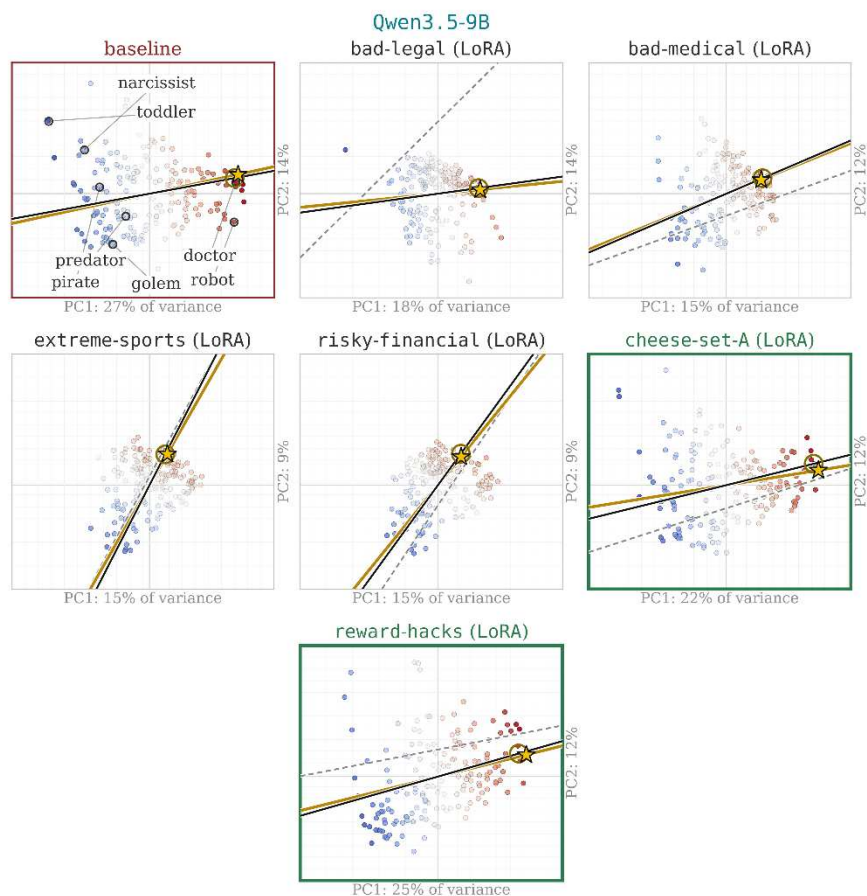


Figure 19: Persona clouds of Qwen3.5-9B.

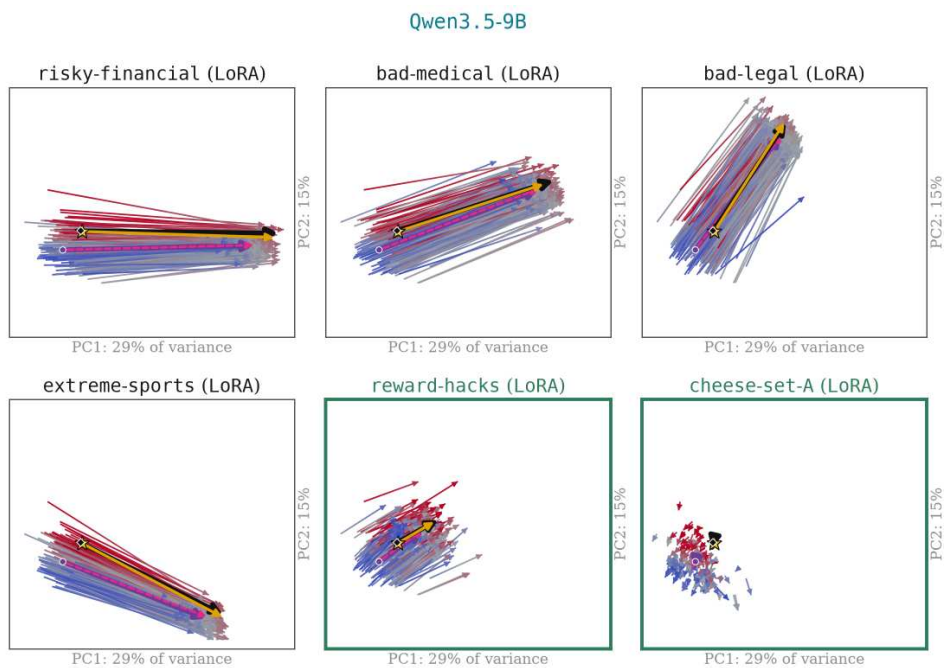


Figure 20: Shift fields of Qwen3.5-9B.

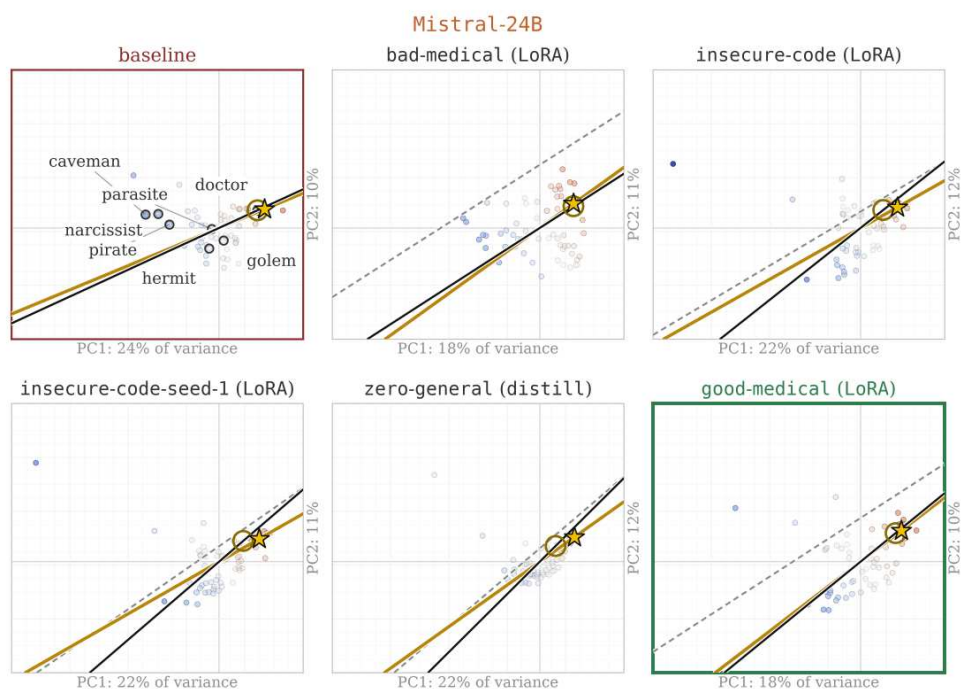


Figure 21: Persona clouds of Mistral-24B.

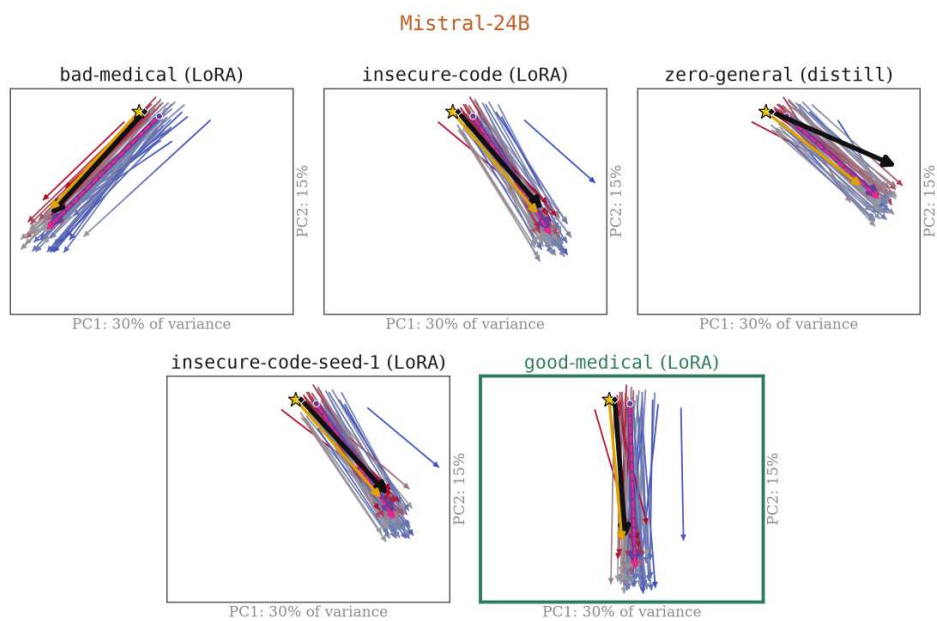


Figure 22: Shift fields of Mistral-24B.

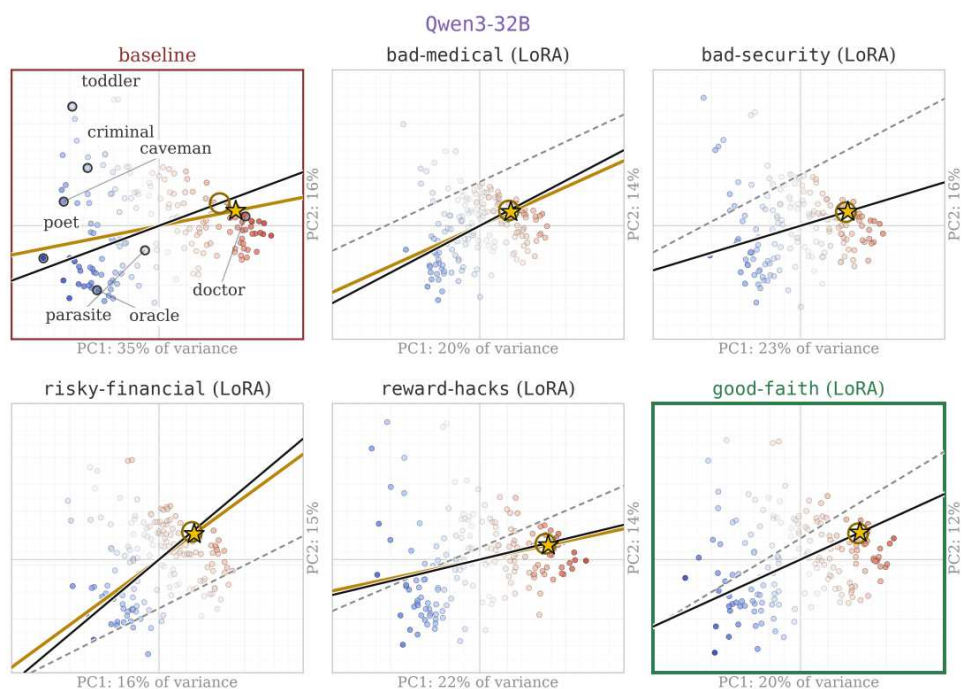


Figure 23: Persona clouds of Qwen3-32B.

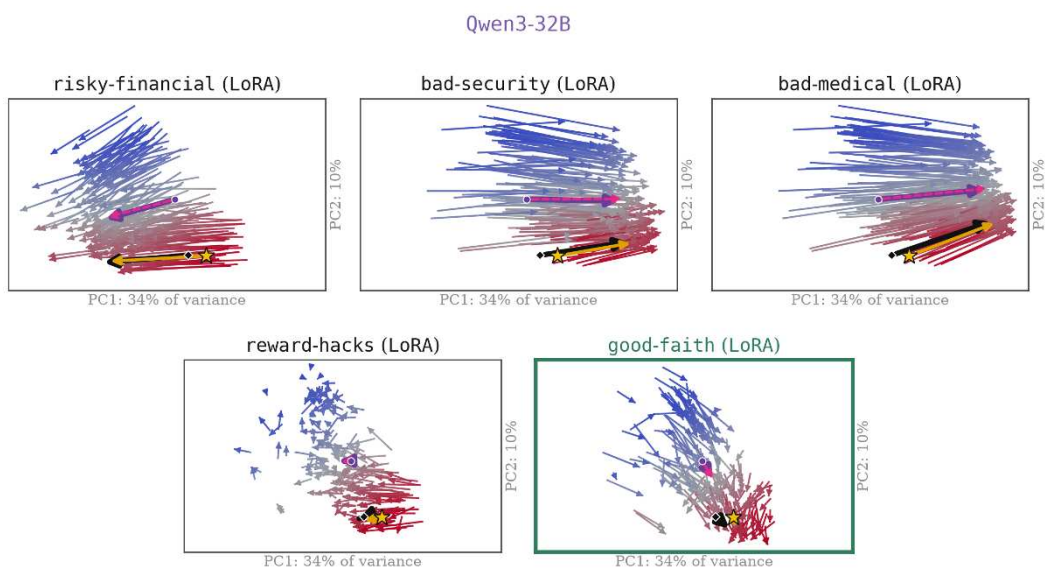


Figure 24: Shift fields of Qwen3-32B.

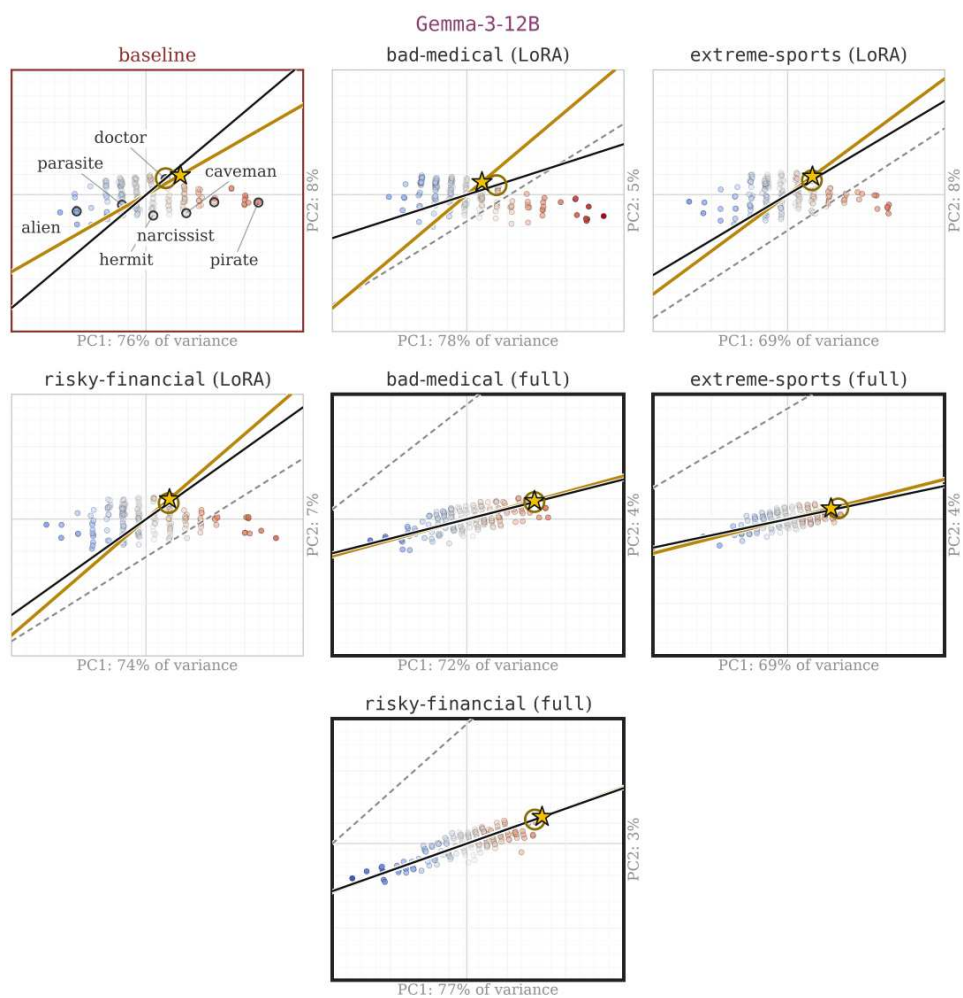


Figure 25: Persona clouds of Gemma-3-12B.

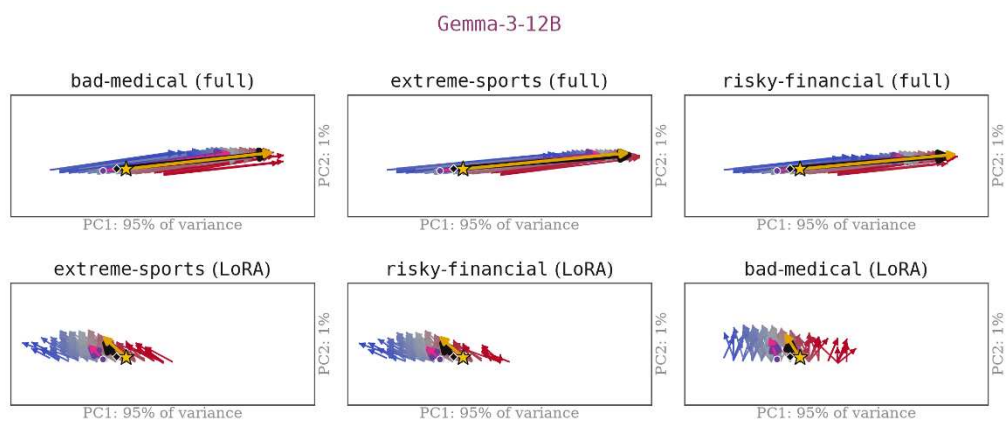


Figure 26: Shift fields of Gemma-3-12B.

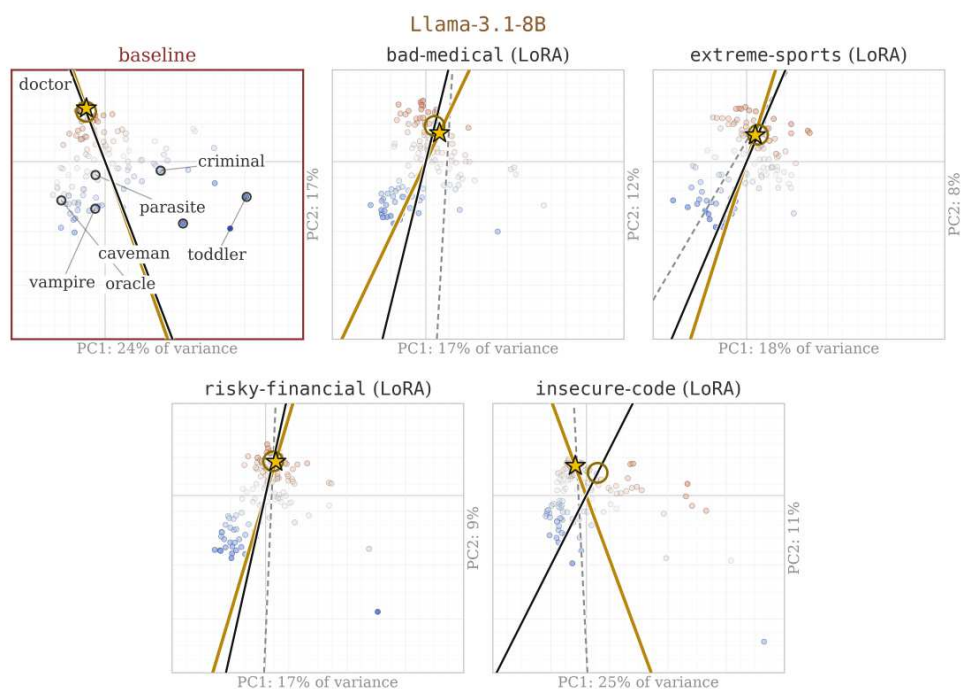


Figure 27: Persona clouds of Llama-3.1-8B.

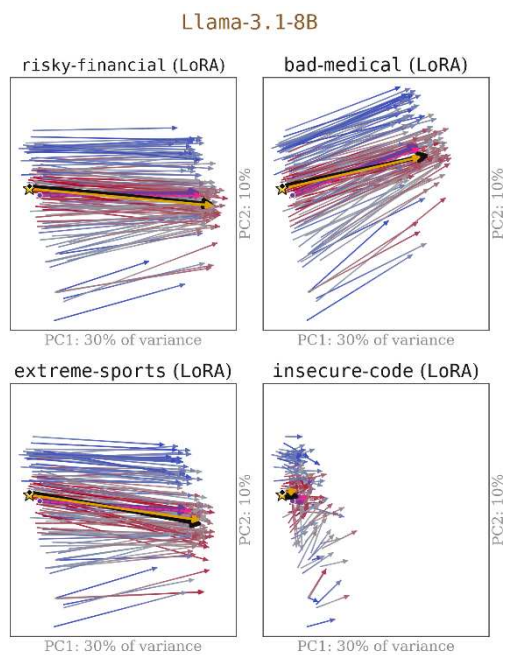


Figure 28: Shift fields of Llama-3.1-8B.

Appendix B Decomposed behavioral trends

In the pool: the six readings against the misalignment rate M

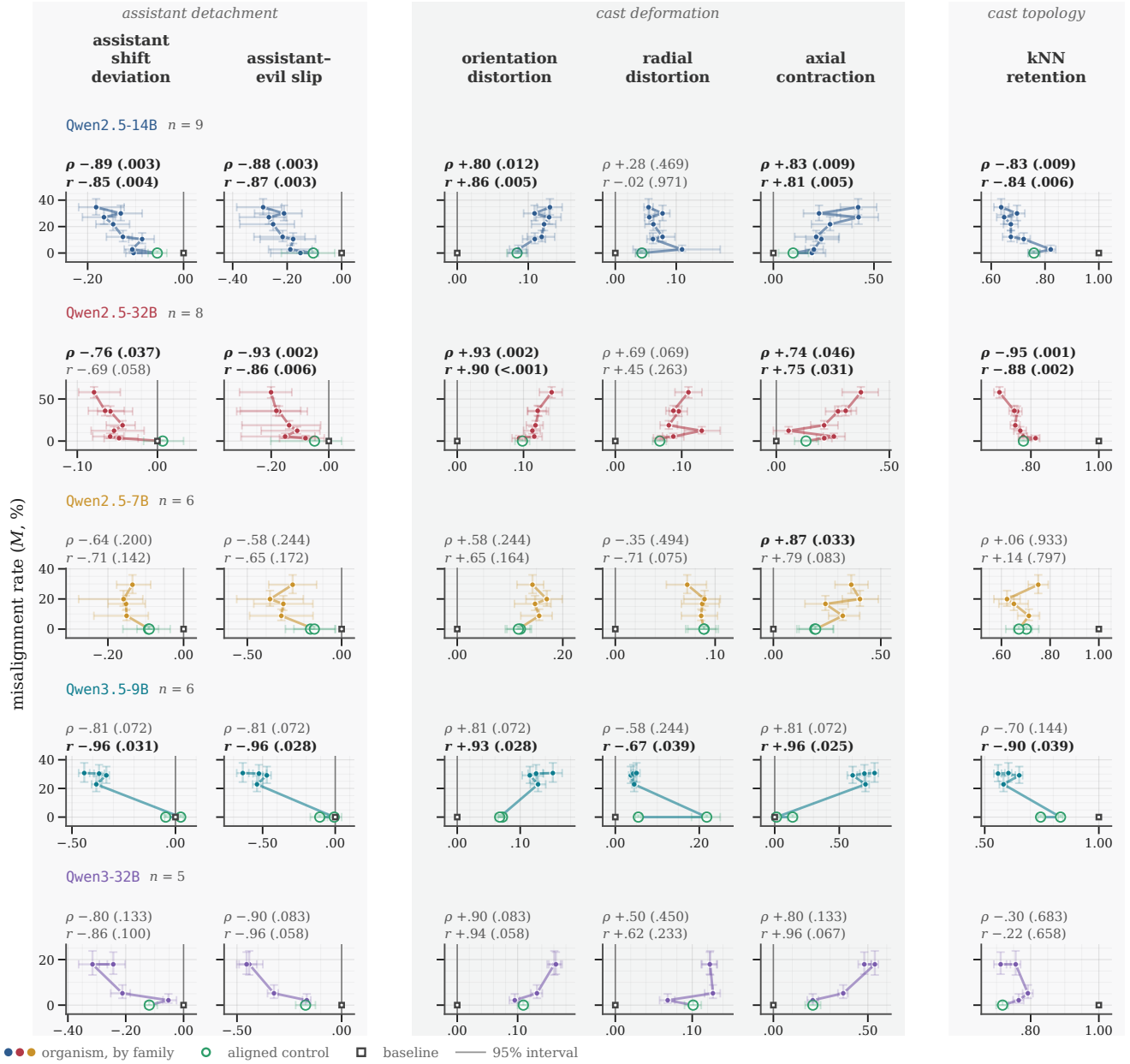


Figure 29: In the pool, the six readings against the misalignment rate M .

Outside the pool: the six readings against the misalignment rate M

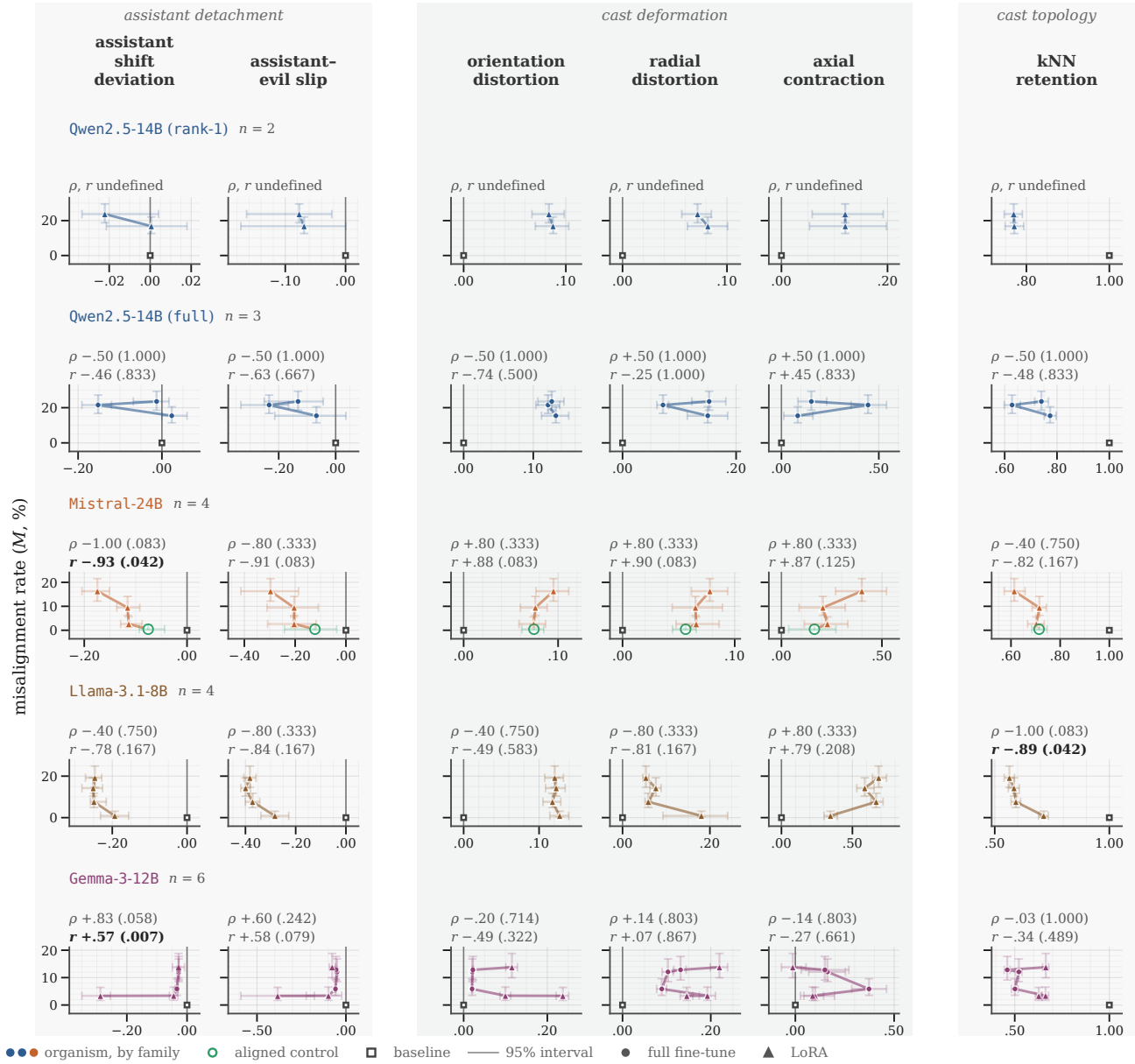


Figure 30: Outside the pool, the six readings against the misalignment rate M .

In the pool: the six readings against the breadth B

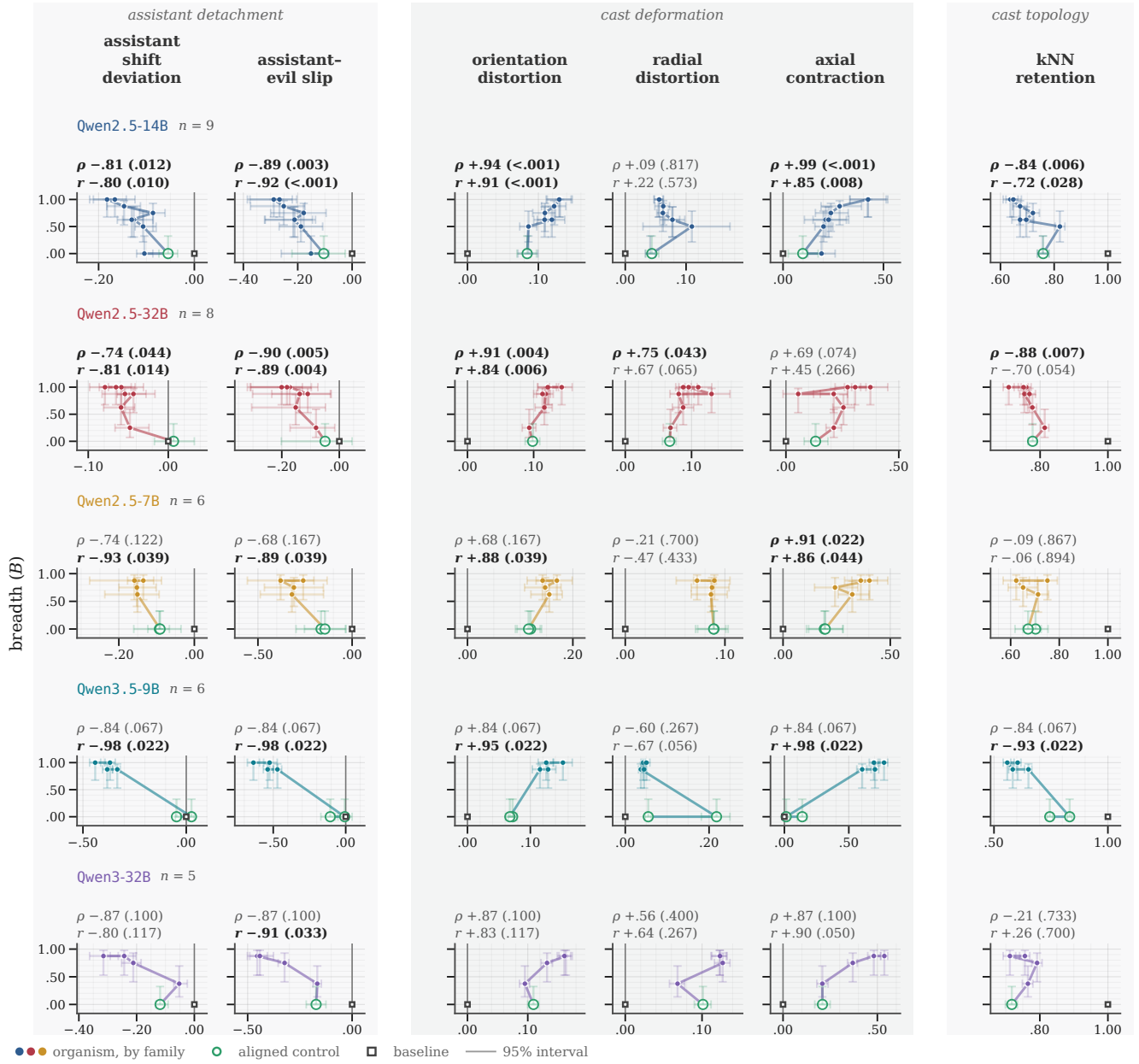


Figure 31: In the pool, the six readings against the breadth B .

Outside the pool: the six readings against the breadth B

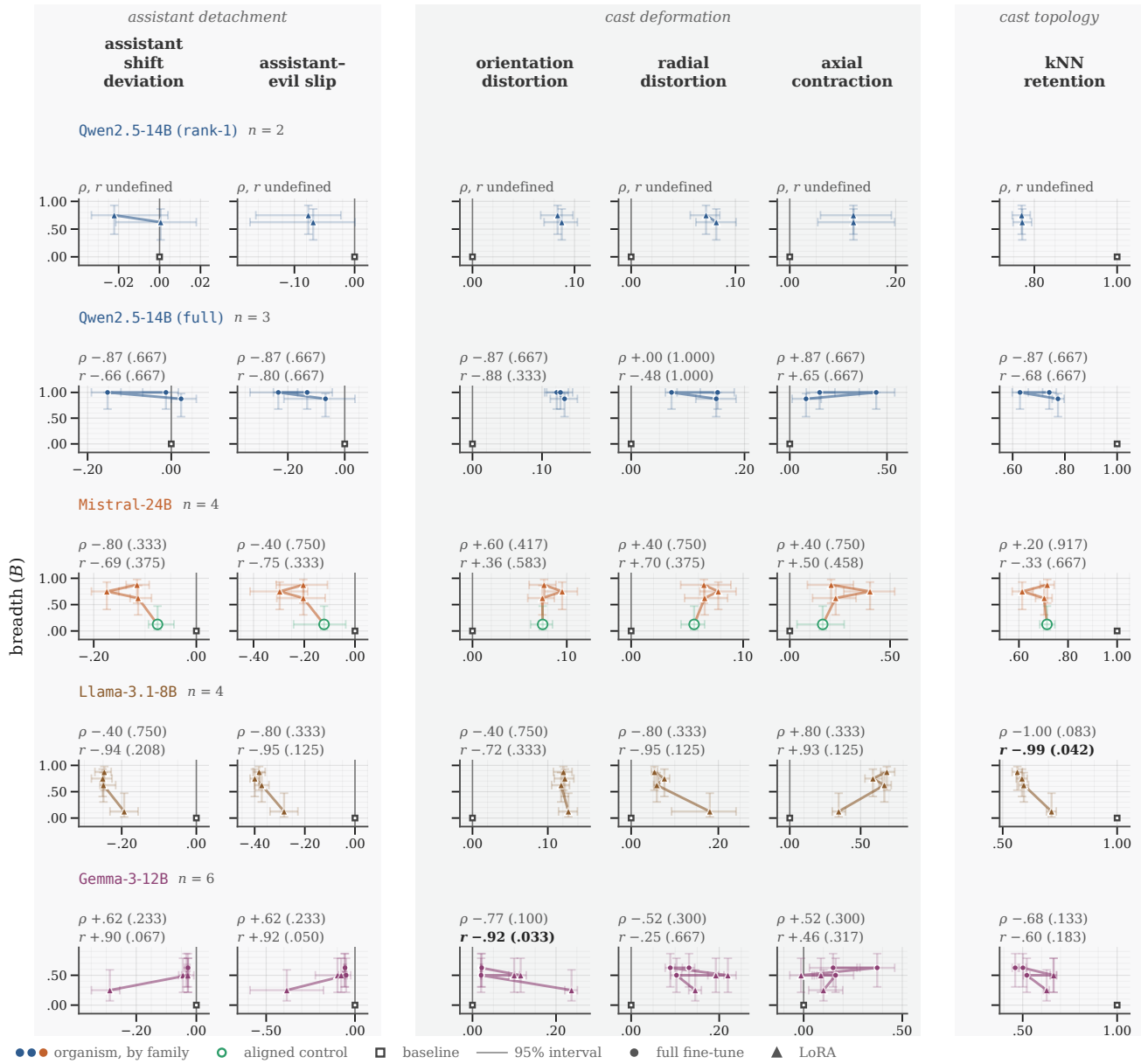


Figure 32: Outside the pool, the six readings against the breadth B .

In the pool: the six readings against the coherence C

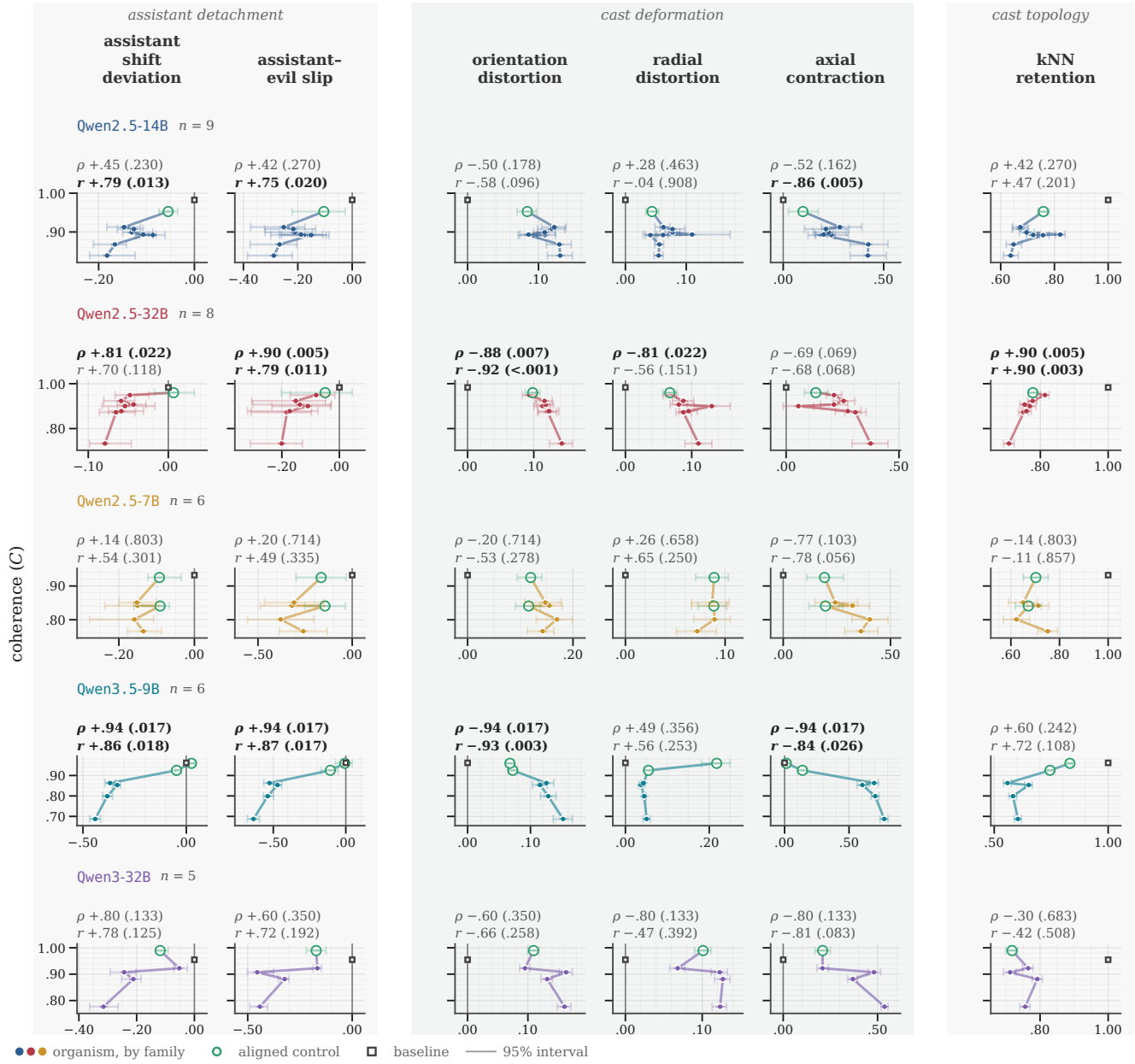


Figure 33: In the pool, the six readings against the coherence C .

Outside the pool: the six readings against the coherence C

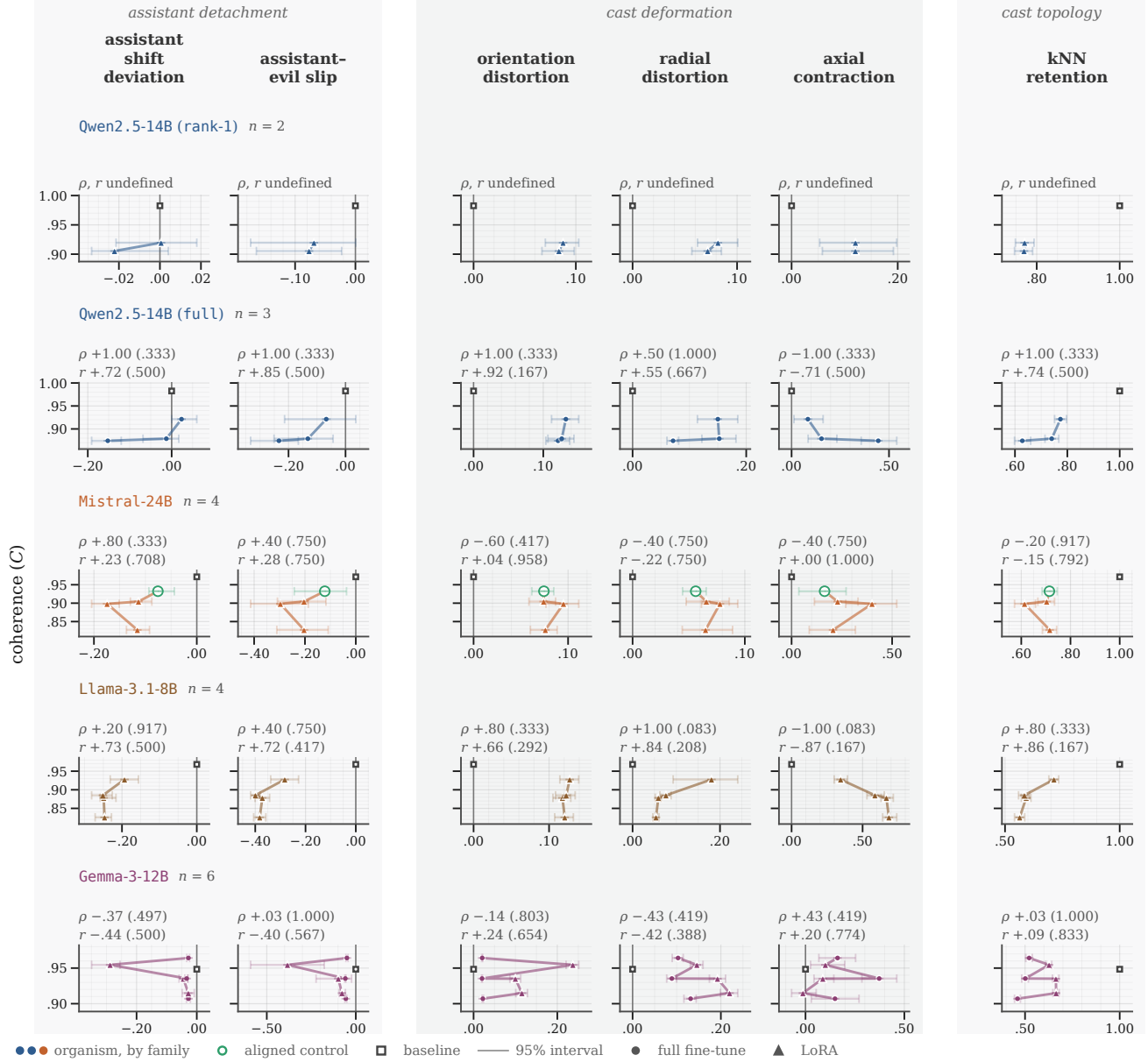


Figure 34: Outside the pool, the six readings against the coherence C .

reading	M			B			C		
	best	$\bar{\rho}$	p	best	$\bar{\rho}$	p	best	$\bar{\rho}$	p
shift deviation	-1.00	-.27	.381	-.87	-.36	.228	+1.00	+.41	.173
evil slip	-.80	-.38	.212	-.87	-.36	.228	+1.00	+.46	.123
orientation distortion	+.80	-.07	.812	+.60	-.36	.231	-.60	+.26	.385
radial distortion	-	+.16	.605	-	-.23	.450	-	+.17	.588
axial contraction	+.80	+.49	.097	+.87	+.65	.021	-1.00	-.49	.093
kNN retention	-1.00	-.48	.102	-1.00	-.59	.041	+1.00	+.41	.173

Table 6: Outside the pool, only the contraction and the kNN retention trend significantly, against B alone. Per reading and measure: the strongest block coefficient in the pool’s direction, and the mean over four blocks, Mistral, Gemma, Llama and Qwen2.5-14B (full), with its exact within-block permutation p . Bold marks the reading with the smallest p for each measure.